

# Macroeconomic Sentiments and Job Search Behavior

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## Abstract

Employed workers who expect the labor market to worsen perceive a greater risk of losing their own jobs and search more intensively while employed. Using survey data, I show that this relationship is driven by workers' fear of job loss rather than their expectations about finding a new job. I interpret this as precautionary on-the-job search: workers look harder for outside offers to protect themselves against the possibility of unemployment. To quantify the mechanism, I embed this behavior in a search-and-matching model that separates precautionary search from ordinary job-ladder search. The model matches the empirical search response, offer receipt, total employer-to-employer mobility, and the share of employer-to-employer moves related to job loss. The calibrated model implies that job-loss beliefs matter for how much employed workers search and for the cyclical nature of that search. The aggregate unemployment effect is modest: in the SCE core calibration, endogenous employed search lowers recession unemployment by about 0.05 percentage points. The contribution is therefore less a stabilization result than a disciplined decomposition of employed search into job-ladder and precautionary components.

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# 1 Introduction

Do households adjust their job search effort before labor market conditions worsen? Recessions make job search significantly harder due to the scarcity of jobs, an increasing pool of job seekers and the increased risk of unemployment. If households foresee an economic downturn in the future, they may also perceive a higher probability of losing their own job and search while still employed to mitigate this risk. Using individual-level survey data from the United States, I show that employed workers who are pessimistic about the economy also expect higher risk of job loss and search more on the job. I then use these facts to discipline a search-and-matching model and quantify what they imply for search behavior and unemployment flows.

This paper makes three contributions. First, I show that pessimism about the economy and perceived job-loss risk strongly predict an increase in search intensity among employed workers. Second, I show that the response is consistent with precautionary search. It is tied to perceived job-loss risk and job-security motives. Third, I quantify the mechanism in a Diamond–Mortensen–Pissarides model with endogenous on-the-job search ([Mortensen and Pissarides, 1994](#)).<sup>1</sup>

Using the Survey of Consumer Expectations (SCE), a monthly panel that elicits probabilistic expectations alongside detailed measures of on-the-job search ([Armantier et al., 2017](#)), I document that a one standard deviation increase in macro pessimism is associated with roughly 24 additional minutes of search per week among employed workers.<sup>2</sup> This association operates largely through personal labor-market expectations. Workers who are more pessimistic about aggregate unemployment report higher job-loss risk, while their expected job-finding probability conditional on job loss moves much less. The same belief-to-risk-to-search pattern also appears in the ECB Consumer Expectations Survey, providing an external check on the mechanism. Three pieces of evidence point to a precautionary motive behind this mechanism. Among employed workers who search, higher perceived job-loss risk predicts a higher probability of citing job security as a search motive. The search response is also strongest among workers who report both high job-loss risk and weak re-employment prospects, the case in which an outside offer is most useful. Finally, perceived job-finding probability enters negatively: workers who expect to find work

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<sup>1</sup>The baseline model treats employed search as a way to build outside options before job loss occurs. I then extend the model so the same offer technology can also generate ordinary job-ladder moves. This allows the model to distinguish the insurance value of outside options from the broader volume of employer-to-employer mobility.

<sup>2</sup>The paper primarily measures macro pessimism as the subjective probability that unemployment will be higher in 12 months. I use other measures of macro expectations such as credit access, interest rates, house prices, government debt and stock prices as robustness.

quickly if separated search *less*. This pattern is hard to reconcile with a pure job-ladder story, but it follows naturally if search partly insures against unemployment.

Reported job-loss probabilities are also predictive of later separations. Workers who assign a higher probability to losing their job are more likely to separate over the subsequent three months, and the relationship is monotone across belief quintiles. This is consistent with the idea that some workers have advance information about job loss (Hendren, 2017).<sup>3</sup> The same beliefs that predict search also forecast later separations. I use these facts to motivate and discipline the model.

To quantify the mechanism, I build a Diamond–Mortensen–Pissarides model with two aggregate states: expansions and recessions. Employed workers face a risk of job loss and choose how much to search while still employed. Search generates outside offers, with the offer probability depending on search effort and labor-market tightness. The key distinction is between the job-loss risk workers actually face and the job-loss risk they perceive. Workers choose search using their perceived risk: a worker who thinks job loss is more likely values an outside offer more and searches harder. Realized flows into unemployment, however, are governed by actual separations. In the baseline model, an outside offer is valuable because it can keep a displaced worker from entering unemployment. I then add a job-ladder extension, where the same offers can also generate ordinary employer-to-employer moves when the current job survives. This lets the model separate the insurance value of an offer from the broader job-ladder value of search.

The calibrated model delivers three results. First, workers search about 26 percent more in recessions than in expansions. If perceived job-loss risk is held fixed across aggregate states, search becomes procyclical: lower tightness makes offers harder to get and discourages effort. Beliefs are therefore central for matching the countercyclicality of employed search. Second, in the baseline outside-option model, precautionary search meaningfully dampens unemployment. Holding search fixed at its expansion level raises recession unemployment by 0.60 percentage points, so the search response offsets about 16 percent of the recessionary rise in unemployment. The mechanism is simple: workers who search more are more likely to have an offer in hand when job loss occurs, and some of those offers turn potential unemployment spells into job-to-job moves. Third, the job-ladder extension limits how large this aggregate channel can be. Once the model also matches ordinary E-to-E mobility and the share of E-to-E moves related to job loss, the unemployment effect falls to 0.1 percentage points, about 2.5 percent of the recessionary

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<sup>3</sup>Advance information and precautionary behavior are distinct. I show that workers who perceive higher risk adjust search in a way that is consistent with self-insurance, especially when they are pessimistic about re-employment.

rise. The behavioral result remains: perceived job-loss risk is central for why employed search rises in recessions. The aggregate unemployment effect depends on how much of that search actually insures job loss.

Finally, a belief-shock counterfactual isolates this decision margin. Holding actual labor-market conditions fixed, workers' perceived job-loss risk is increased. Workers search more because the outside offer is more valuable as insurance. Offer arrival rates rise, and employment-to-unemployment transitions fall slightly because more separated workers have offers available.

The paper relates to three strands of work. First, it contributes to research on subjective expectations and labor-market behavior. Workers' beliefs about labor-market prospects are informative and behaviorally relevant: employed workers have advance information about job loss ([Hendren, 2017](#)), and unemployed job seekers' beliefs predict outcomes but can be biased ([Spinnewijn, 2015](#); [Mueller et al., 2021](#)). I add evidence on a different margin: employed workers' perceived job-loss risk predicts how much they search before any separation occurs.

Second, the paper relates to the on-the-job search and job-ladder literature. Standard models emphasize search for better jobs and higher wages ([Burdett and Mortensen, 1998](#); [Postel-Vinay and Robin, 2002](#); [Menzio and Shi, 2011](#); [Moscarini and Postel-Vinay, 2018](#)). Recent work shows that search is quantitatively important and often cyclical. [Bransch \(2021\)](#) studies the search intensity of unemployed workers over the business cycle, while [Faberman et al. \(2022\)](#) document that on-the-job search is pervasive and effective. [Ahn and Shao \(2017\)](#) show that employed search is countercyclical and related to precautionary concerns, and [Simmons \(2024\)](#) studies precautionary on-the-job search before job loss and its implications for mobility and insurance. Relative to this work, I use direct subjective job-loss probabilities to separate perceived separation risk from perceived job-finding prospects, use stated search motives to discipline the interpretation, and quantify the implied search and unemployment-flow margins in a model that also matches ordinary job-to-job mobility.

Third, the paper contributes to work on expectations and business-cycle fluctuations. Survey expectations matter for household decisions, and disagreement in macro beliefs is widespread ([Mankiw et al., 2003](#); [Coibion et al., 2018](#)). In labor markets, macro perceptions can affect personal expectations and behavior ([Roth and Wohlfart, 2019](#); [Carroll and Dunn, 1997](#)). [Pilososop and Ryngaert \(2024\)](#) show that inflation expectations affect employed search through nominal wages. Here, perceived job-loss risk changes search because outside offers help workers avoid unemployment. This also connects the paper to work on search effort over the cycle. [Mukoyama et al. \(2018\)](#) show that countercyclical search among

the unemployed affects unemployment fluctuations; this paper studies a related inflow margin among employed workers, while finding a more modest aggregate unemployment effect.

The rest of the paper is organized as follows. Section 2 documents the empirical relationship between perceived separation risk and on-the-job search. Section 3 develops a search model with job-ladder and precautionary motives. Section 4 describes the calibration and model fit. Section 5 reports the quantitative implications. Section 6 concludes.

## 2 Perceived Separation Risk and On-the-Job Search

This section uses individual-level data from the Survey of Consumer Expectations (SCE) to study how employed workers' labor-market expectations relate to on-the-job search. Workers who report higher job-loss risk search more while employed. Several facts documented in this section point to a precautionary motive: reported job-loss risk predicts subsequent separations, predicts job-security search motives, and is most important when perceived re-employment prospects are weak.

The SCE analysis proceeds in four steps. Section 2.2 relates expectations to search. Section 2.3 shows that macroeconomic pessimism is more closely related to personal job-loss expectations than to perceived job-finding prospects. Section 2.4 shows that reported job-loss probabilities forecast realized separations. Section 2.5 uses stated search motives and interaction patterns to distinguish precautionary search from other reasons for on-the-job search. After robustness checks, Section 2.7 asks whether the same belief-to-risk-to-search link appears in the ECB Consumer Expectations Survey.

### 2.1 The Survey of Consumer Expectations

I use the Survey of Consumer Expectations (SCE), a nationally representative monthly internet-based panel conducted by the Federal Reserve Bank of New York since 2013. Each month approximately 1,300 household heads complete the survey; respondents typically remain in the panel for 12 consecutive months. The SCE has two components relevant to this paper. The *core monthly survey* collects employment status and probabilistic expectations—elicited on a 0–100 scale—about aggregate and personal labor market outcomes. The *Labor Market Module* (LMM) is administered to the same respondents every four months, so each participant completes it up to three times during their panel tenure. The LMM collects detailed information on job search behavior. For further details on the SCE, see [Armantier et al. \(2017\)](#).

**Labor Market Module.** The LMM’s primary search measure asks: “How many hours did you spend in the past 7 days looking or applying for jobs?” This question captures the *intensive margin* of on-the-job search. The module also records which search activities the respondent undertook (browsing postings, applying online, networking, contacting employers directly) and, for the mechanism tests in Section 2.5, the respondent’s stated reasons for searching. The analysis sample consists of 4,226 employed worker-months with non-missing search hours, spanning March 2014 through December 2023.<sup>4</sup>

**Expectation variables.** The empirical analysis relies on four probabilistic expectation variables from the SCE core survey, each elicited on a 0–100 scale as a point estimate rather than a full density forecast. *Macro pessimism* is the respondent’s subjective probability that “12 months from now the unemployment rate will be higher than it is now” (sample mean: 39.4%, SD: 23.9 pp). *Job-loss probability* is the percent chance the respondent assigns to involuntarily losing their current job in the next 12 months (mean: 21.3%, SD: 23.4 pp). *Voluntary separation probability* is the percent chance of voluntarily leaving the current job in the next 12 months (mean: 40.2%, SD: 30.4 pp). *Job-finding probability conditional on job loss* is the percent chance of finding a new job within three months if the current job is lost (mean: 57.1%, SD: 30.0 pp). Expected inflation is also collected in the core survey and enters the regressions as a control. Because the survey elicits point estimates, workers with the same reported probability may differ in subjective uncertainty; this heterogeneity is not observed.

Appendix Table A1 reports summary statistics. The average employed worker in the LMM reports approximately 3.8 hours per week of search (se: 4.8 hours/week).

## 2.2 Perceived risk and search intensity

This subsection discusses how expectations relate to search intensity. A worker’s search effort is a forward-looking decision variable and it is naturally influenced by their expectations about the macroeconomy as well as their expectations about their personal circumstances. I condition on a rich set of controls and fixed effects, so the estimates compare workers facing the same time and location environment. The baseline specification regresses weekly search hours on the worker’s subjective probability that unemployment

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<sup>4</sup>The full SCE Labor Market Module contains approximately 12,000 respondent-wave observations over this period; the analysis sample restricts to employed workers with non-missing search hours and expectation variables.

will be higher in 12 months:

$$\text{SearchHours}_{ist} = \beta E_{it}[\Delta \Pr(U_{t+12} \geq 0)] + \mathbf{X}'_{it}\gamma + \mu_s + \lambda_t + \varepsilon_{ist}, \quad (1)$$

where  $E_{it}[\Delta \Pr(U_{t+12} \geq 0)]$  is scaled to  $[0, 1]$ ,  $\mathbf{X}_{it}$  includes demographic controls, and  $\mu_s$  and  $\lambda_t$  denote state and time fixed effects. Standard errors are clustered at the individual level. Figure 1 (Panel A) displays the conditional relationship between macro pessimism and search hours after residualizing both variables with respect to controls and fixed effects.



Notes: Each panel plots binned means of weekly search hours against a single expectation measure after residualizing both variables on the full set of controls, state fixed effects, and time fixed effects. Expectations are scaled to  $[0, 1]$  and search hours are winsorized at the 5th and 95th percentiles. Sample: employed respondents in the SCE Labor Market Module.

**Figure 1.** Binscatter: expectations vs. search hours

Column (1) of Table 1 reports the corresponding coefficient. A one-standard-deviation increase in macro pessimism (24 percentage points) is associated with about 24 additional minutes of job search per week, roughly 10 percent of mean search. The estimate changes little when I include expectations about other macro variables, including inflation, stock prices, credit access, house prices, interest rates, and government debt. Figure 1 (Panel A) shows the corresponding conditional binscatter.

Aggregate beliefs are informative, but workers also hold beliefs about their own labor market prospects. Columns (2)–(5) of Table 1 therefore incorporate personal expectations about job finding and separation. Two results stand out: perceived *job-loss probability* strongly predicts higher search intensity, while perceived *job-finding probability conditional on job loss* enters with the opposite sign. Workers who expect to find a job quickly if separated search *less*. These signs are consistent with precautionary search: an outside offer is more valuable when job-loss risk is high and unemployment would be costly. Voluntary separation expectations are included as an additional control and enter positively, consistent with a job-ladder margin;<sup>5</sup> the central mechanism in this paper is the perceived

<sup>5</sup>The job-ladder literature emphasizes that employed workers search to climb toward better-paying jobs; see Postel-Vinay and Robin (2002) and Moscarini and Postel-Vinay (2018) for theory and evidence.



job-loss channel.

**Table 1.** Expectations and On-the-Job Search Intensity

|                                                                                   | (1)<br>Baseline     | (2)<br>+ Separation | (3)<br>+ P(find)     | (4)<br>+ P(offer)    | (5)<br>Interaction  |
|-----------------------------------------------------------------------------------|---------------------|---------------------|----------------------|----------------------|---------------------|
| $E_t(\Delta \Pr(U_t + 12 \geq 0))$                                                | 1.607***<br>(0.372) | 0.953***<br>(0.356) | 0.953***<br>(0.355)  | 0.872**<br>(0.370)   | 0.928***<br>(0.352) |
| $E_t(\Pr(\text{Lose Job}_t + 12))$                                                |                     | 2.046***<br>(0.425) | 1.836***<br>(0.429)  | 1.629***<br>(0.441)  | 2.682***<br>(0.831) |
| $E_t(\Pr(\text{Leave Job}_t + 12))$                                               |                     | 1.567***<br>(0.330) | 1.884***<br>(0.333)  | 1.313***<br>(0.361)  | 1.890***<br>(0.334) |
| $E_t(\Pr(\text{Find} \text{Lose}_t + 3))$                                         |                     |                     | -1.156***<br>(0.294) | -1.568***<br>(0.298) | -0.877**<br>(0.365) |
| $E_t(\Pr(\text{Offer}_t + 3))$                                                    |                     |                     |                      | 0.019***<br>(0.003)  |                     |
| $E_t(\Pr(\text{Lose}_t + 3)) \times$<br>$E_t(\Pr(\text{Find} \text{Lose}_t + 3))$ |                     |                     |                      |                      | -1.479<br>(1.214)   |
| $E_t(\pi_t + 12)$ (dist. mean)                                                    | -0.025<br>(0.021)   | -0.025<br>(0.022)   | -0.028<br>(0.022)    | -0.024<br>(0.023)    | -0.029<br>(0.022)   |
| Time FE                                                                           | Yes                 | Yes                 | Yes                  | Yes                  | Yes                 |
| State FE                                                                          | Yes                 | Yes                 | Yes                  | Yes                  | Yes                 |
| Observations                                                                      | 4132                | 4131                | 4130                 | 3922                 | 4130                |
| R-squared                                                                         | 0.125               | 0.148               | 0.152                | 0.163                | 0.153               |

Notes: Standard errors in parentheses. Dependent variable is weekly job search hours (winsorized). Sample is employed workers. All specifications include state and time fixed effects, are weighted by survey weights, and control for age polynomial, education, race, gender, marital status, and household income. Standard errors are clustered by individual. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

After conditioning on personal expectations, the macro pessimism coefficient falls but remains positive. This decline suggests that macro pessimism predicts search in part because it is related to workers' own perceived job-loss risk. The remaining coefficient may capture other aggregate risks not covered by the SCE's personal expectation questions.

To make the comparison more discrete, Table 2 repeats the same exercise using indicators for whether each of the four core expectations lies above the sample median. This specification compares workers with relatively high beliefs to those at or below the median rather than tracing marginal responses to a one-unit change in reported probabilities.

The indicator specification delivers the same qualitative pattern. Median search among employed workers is about 2 hours per week, so these differences are economically mean-



**Table 2.** Expectations and On-the-Job Search Intensity: Above-Median Indicators

|                                                                                                                                               | (1)<br>Baseline     | (2)<br>+ Separation | (3)<br>+ P(find)     | (4)<br>+ P(offer)    | (5)<br>Interaction  |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|----------------------|----------------------|---------------------|
| Above median:<br>$E_t(\Delta \Pr(U_{t+12} \geq 0))$                                                                                           | 0.765***<br>(0.174) | 0.519***<br>(0.165) | 0.520***<br>(0.165)  | 0.538***<br>(0.170)  | 0.512***<br>(0.164) |
| Above median:<br>$E_t(\Pr(\text{Lose Job}_{t+12}))$                                                                                           |                     | 0.673***<br>(0.170) | 0.588***<br>(0.170)  | 0.496***<br>(0.174)  | 0.860***<br>(0.248) |
| Above median:<br>$E_t(\Pr(\text{Leave Job}_{t+12}))$                                                                                          |                     | 0.977***<br>(0.176) | 1.075***<br>(0.176)  | 0.751***<br>(0.193)  | 1.077***<br>(0.177) |
| Above median:<br>$E_t(\Pr(\text{Find} \text{Lose}_{t+3}))$                                                                                    |                     |                     | -0.595***<br>(0.169) | -0.836***<br>(0.170) | -0.311<br>(0.229)   |
| $E_t(\Pr(\text{Offer}_{t+3}))$                                                                                                                |                     |                     |                      | 0.019***<br>(0.003)  |                     |
| Above median:<br>$E_t(\Pr(\text{Lose Job}_{t+12})) \times$<br>Above median:<br>$E_t(\Pr(\text{Find} \text{Lose}_{t+3}))$<br>$E_t(\pi_{t+12})$ | -0.023<br>(0.021)   | -0.024<br>(0.022)   | -0.027<br>(0.022)    | -0.022<br>(0.023)    | -0.027<br>(0.022)   |
| Time FE                                                                                                                                       | Yes                 | Yes                 | Yes                  | Yes                  | Yes                 |
| State FE                                                                                                                                      | Yes                 | Yes                 | Yes                  | Yes                  | Yes                 |
| Observations                                                                                                                                  | 4132                | 4131                | 4130                 | 3922                 | 4130                |
| R-squared                                                                                                                                     | 0.125               | 0.142               | 0.145                | 0.158                | 0.146               |

Notes: Dependent variable is weekly job search hours (winsorized). The four core expectations from Table 1 are replaced by indicators for responses strictly above the sample median in the employed sample with non-missing search hours; coefficients therefore measure search-hour differences relative to workers at or below the median. Column (4) retains the continuous expected-offer control, and Column (5) interacts the above-median job-loss and job-finding indicators. All specifications include time and state fixed effects, are weighted by survey weights, and control for age polynomial, education, race, gender, marital status, and household income. Standard errors clustered by individual. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

ingful. Workers with above-median macro pessimism search about 0.8 hours more per week, and workers with above-median perceived job-loss risk search about 0.9 hours more per week. Above-median voluntary-separation beliefs are also associated with more search, while above-median job-finding beliefs are associated with less search in the additive specifications. The interaction column makes the precautionary interpretation clearer as among workers with below-median job-finding beliefs, moving above the median in perceived job-loss risk raises search by about 0.9 hours per week; when job-finding beliefs are also above the median, this incremental effect is smaller by about 0.60 hours. In other words, the search premium associated with high perceived separation risk is concentrated among workers who are not confident they can quickly find a new job after separation.

### 2.3 From macro pessimism to personal separation risk

How do aggregate expectations map into personal beliefs that are directly relevant for search? Table 3 investigates this link by regressing personal expectation variables on macro unemployment pessimism, controlling for demographics and fixed effects. The estimates reveal a sharp asymmetry: macroeconomic pessimism maps strongly into perceived separation risk but has essentially no association with perceived job-finding probability conditional on job loss.

Workers who are more pessimistic about aggregate unemployment report substantially higher perceived job-loss probability: a worker at the 75th percentile of macro pessimism perceives roughly 4 percentage points higher separation risk than a worker at the 25th percentile. Macro pessimism also correlates with voluntary separation expectations, but the key asymmetry is that it has essentially no relationship with perceived job-finding probability conditional on job loss. This pattern is central for the paper’s mechanism: it implies that aggregate pessimism is linked primarily to the perceived *inflow risk* into unemployment rather than perceived duration once unemployed.

Consistent with this pattern, controlling for personal expectations reduces the macro pessimism coefficient in the search regression by about 40 percent. A product-of-coefficients calculation attributes most of this reduction to job-loss expectations, with a secondary role for voluntary-separation beliefs. In the remainder of the paper, the quantitative model uses the job-loss-risk relationship to evaluate the aggregate implications of precautionary on-the-job search.

**Table 3.** Macro Expectations and Individual Labor Market Expectations

|                                    | (1)                   | (2)                   | (3)                    |
|------------------------------------|-----------------------|-----------------------|------------------------|
|                                    | P(Lose Job)           | P(Leave Job)          | P(Find Job)            |
| $E_t(\Delta \Pr(U_{t+12} \geq 0))$ | 0.1693***<br>(0.0062) | 0.1362***<br>(0.0080) | -0.0034<br>(0.0111)    |
| $E_t(\pi_{t+12})$                  | 0.0001<br>(0.0003)    | -0.0004<br>(0.0003)   | -0.0013***<br>(0.0004) |
| Observations                       | 92634                 | 92650                 | 92824                  |
| R-squared                          | 0.075                 | 0.060                 | 0.068                  |

Notes: Dependent variables are subjective probabilities (0–1). Regressor is macro unemployment pessimism,  $\Delta \Pr(U_{t+12} \geq 0)$ , with expected inflation as a control. Sample: employed workers; time and state fixed effects plus demographics; weighted by survey weights; standard errors clustered by individual. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## 2.4 Do beliefs predict realized separations?

A key concern is whether reported job-loss probabilities contain information about future outcomes or instead capture noise or broad sentiment. The SCE’s panel structure allows a direct check: do workers who report higher job-loss probability actually separate at higher rates?

Table 4 shows that perceived job-loss risk predicts subsequent separations. A ten percentage point increase in perceived job-loss probability is associated with a 1.3 percentage point higher realized separation rate over the subsequent three months. Although the belief horizon is 12 months, the measure forecasts near-term separations, consistent with persistent differences in separation risk across workers and jobs.<sup>6</sup> Appendix Table A11 shows that this relationship persists across multiple horizons (1–9 months and censored 12-month outcomes), and Appendix Table A12 shows that it remains positive and significant under both broad non-employment and strict unemployment outcome definitions.

Figure 2, Panel A, provides a nonparametric view. Sorting workers into quintiles of subjective job-loss expectations yields an upward pattern: workers in the top quintile separate at substantially higher rates than workers in the bottom quintile. This evidence is consistent with Stephens (2004) and Mueller et al. (2021), who document that subjective job-loss expectations in household surveys predict actual displacement.

<sup>6</sup>The 12-month belief horizon exceeds the 3-month outcome window, but persistent heterogeneity in separation risk across workers implies that beliefs should predict near-term separations even if the stated horizon is longer.

**Table 4.** Validation: Subjective Job Loss Expectations Predict Separations (Q10-based status)

|                                 | (1)                   | (2)                   | (3)                   | (4)                   | (5)                   |
|---------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                 | None                  | Controls              | Time FE               | State + Time FE       | State×Time FE         |
| $E_t P(\text{Lose Job}_{t+12})$ | 0.1547***<br>(0.0129) | 0.1320***<br>(0.0121) | 0.1314***<br>(0.0120) | 0.1307***<br>(0.0118) | 0.1290***<br>(0.0107) |
| Controls                        | No                    | Yes                   | Yes                   | Yes                   | Yes                   |
| Fixed Effects                   | None                  | None                  | Time                  | State, Time           | State×Time            |
| Observations                    | 56412                 | 55993                 | 55993                 | 55832                 | 55832                 |
| R-squared                       | 0.021                 | 0.049                 | 0.056                 | 0.059                 | 0.205                 |

Notes: Job-loss expectations are scaled to  $[0, 1]$ , so coefficients are percentage-point changes in separation probability. Status is constructed from Q10 dummies: employed/attached = FT, PT, leave; unemployed = want work, laid off; NILF = disabled, retiree, student, homemaker. Exact-horizon outcomes require observation at  $t + h$  (xtset + F#.). Regressions are weighted by survey weights; SEs clustered by individual.

## 2.5 Mechanism: Evidence of a precautionary motive

The results so far show that perceived job-loss risk predicts search intensity. I now ask whether this relationship reflects precautionary motives—searching to secure job continuity in the event of separation—rather than a generic tendency to search.

**Stated search motives.** The SCE Labor Market Module asks employed workers who report searching to indicate why they are looking for a new job. Table 5 uses this information to test a direct implication of precautionary search: if higher perceived job-loss risk induces search for insurance, it should raise the likelihood that workers cite *job security* as their motive.

Workers with higher perceived job-loss probability are substantially more likely to cite job security as a reason for searching. Voluntary separation expectations, by contrast, enter with the opposite sign: workers who expect to leave voluntarily are *less* likely to cite job security and more likely to cite opportunity-related motives (better pay, advancement). This difference helps separate job-loss expectations from a general desire to move jobs. Finally, conditional on personal expectations, macro pessimism has no independent association with citing job security, suggesting that the security motive operates primarily through beliefs about one’s own job-loss risk.

These patterns help distinguish precautionary search from a generic propensity to search: the same beliefs that predict higher search intensity also predict that workers attribute their search to employment stability concerns.

If workers search to insure against job loss, the value from searching while employed

**Table 5.** Direct Evidence: Why Are Employed Workers Searching?

|                                           | (1)<br>Searching for Job Security | (2)<br>Ladder / Upgrade Search |
|-------------------------------------------|-----------------------------------|--------------------------------|
| $E_t(\text{Pr}(\text{Lose Job}_{t+12}))$  | 0.588**<br>(0.071)                | -0.271***<br>(0.068)           |
| $E_t(\text{Pr}(\text{Leave Job}_{t+12}))$ | -0.145***<br>(0.048)              | 0.040<br>(0.049)               |
| $E_t(\Delta \text{Pr}(U_{t+12} \geq 0))$  | -0.026<br>(0.061)                 | 0.054<br>(0.059)               |
| Time FE                                   | Yes                               | Yes                            |
| State FE                                  | Yes                               | Yes                            |
| Observations                              | 707                               | 707                            |
| R-squared                                 | 0.290                             | 0.211                          |

Notes: Column 1 equals 1 if the worker reports searching due to job stability concerns or having received a layoff notice. Column 2 equals 1 if the worker reports searching for better pay, advancement, hours, fit, quality of life, commute, or wanting a change. Expectations are scaled to  $[0, 1]$ . Sample: employed workers in the SCE Labor Market Module who report searching. All specifications include time and state fixed effects and demographic controls; weighted by survey weights. Standard errors clustered by individual. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

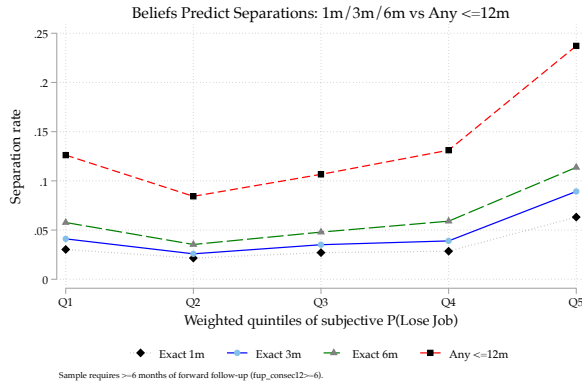
should be highest when perceived job-loss risk is high and expected re-employment prospects are weak. Table 6 tests this by allowing the effect of macro pessimism on search to vary with (i) perceived job-loss risk and (ii) perceived re-employment prospects.<sup>7</sup>

Two patterns stand out. First, the interaction between macro pessimism and perceived job-loss probability is positive and statistically significant. Aggregate pessimism induces a larger increase in search among workers who perceive higher personal separation risk.

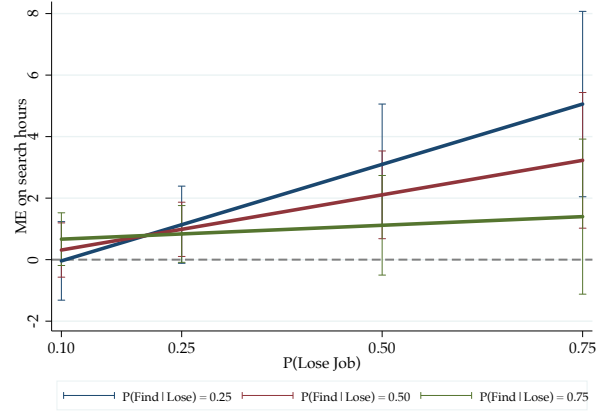
Second, the three-way interaction between macro pessimism, perceived job-loss risk, and perceived job-finding probability is negative and statistically significant. The amplification from job-loss risk is strongest when workers are less confident about finding a job quickly. This is the case in which an outside offer provides the most insurance: high probability of needing it and high expected cost if separation occurs without one.

Formally, the relevant object is the marginal effect of macro pessimism on search, evaluated at different points in the joint distribution of  $(\text{Pr}(\text{Lose}), \text{Pr}(\text{Find}))$ . Figure 2, Panel B, shows that this marginal effect is largest in the high-risk, low-prospect region, where workers feel vulnerable to job loss and pessimistic about rapid re-employment, and smallest when either separation risk is low or re-employment prospects are strong.

<sup>7</sup>Because the expectations are centered, the main coefficients describe effects at average beliefs, and the interaction terms capture how the response varies across workers.



Panel A. Realized separations by perceived job-loss risk



Panel B. Marginal effect of macro pessimism on search

Notes: Panel A sorts employed workers into quintiles of perceived 12-month job-loss probability and plots realized separation rates at several horizons. Panel B reports marginal effects from the interaction specification in Table 6, using centered expectation variables. It shows how the effect of macro unemployment expectations on search hours varies with perceived job-loss risk and perceived job-finding probability. All models include state×time fixed effects and full controls.

**Figure 2.** Mechanism Evidence: Beliefs Predict Risk and Shape Search Responses

## 2.6 Robustness

The baseline estimates document a cross-sectional association between expectations and search. A natural concern is omitted variables: workers in declining industries or weak local labor markets may report both high pessimism and high search intensity because both reflect underlying conditions rather than a behavioral response. The robustness exercises address this and related concerns. Full tables appear in Appendix B.

*Alternative fixed effects.* Replacing separate state and time fixed effects with state×time interactions absorbs all local aggregate variation, so identification relies on differences across workers within the same state-month cell. Under this specification, the macro pessimism coefficient declines modestly but remains statistically significant, and the job-loss expectations coefficient is essentially unchanged (Table A2). With industry×time fixed effects, which absorb sector-level cyclical conditions, the results are similar (Table A3). These specifications reduce the scope for local or sectoral shocks to explain the estimates.

*Functional form and margins.* Using alternative transformations of search hours ( $\log(1 + y)$  and an arc-percent transformation) and Poisson specifications leaves the qualitative results intact (Table A5). The arc-percent specification follows a DHS-style cross-sectional scaling,  $(y - \bar{y}) / (0.5(y + \bar{y}))$ , which avoids relying on the scale-dependent inverse-hyperbolic-sine transformation. At the extensive margin (Table A6), personal separation expectations predict whether a worker searches at all, whereas macro pessimism does not once personal

**Table 6.** Macro Pessimism and Expectations about Personal Outcomes

|                                                                                                                             | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  |
|-----------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                                                                                                             | Baseline             | + Macro×Lose         | + Macro×Find         | + Lose×Find          | Three-way            |
| $E_t(\Delta \Pr(U_{t+12} \geq 0))$                                                                                          | 0.906**<br>(0.412)   | 0.659<br>(0.405)     | 0.493<br>(0.453)     | 0.452<br>(0.451)     | 0.527<br>(0.414)     |
| $E_t(\Pr(\text{Lose Job}_{t+12}))$                                                                                          | 1.841***<br>(0.502)  | 1.616***<br>(0.498)  | 1.645***<br>(0.496)  | 1.694***<br>(0.499)  | 1.592***<br>(0.500)  |
| $E_t(\Pr(\text{Find Job}_{t+3}))$                                                                                           | -1.259***<br>(0.352) | -1.235***<br>(0.352) | -1.220***<br>(0.352) | -1.150***<br>(0.358) | -1.057***<br>(0.360) |
| $E_t(\Delta \Pr(U_{t+12} \geq 0)) \times$<br>$E_t(\Pr(\text{Lose Job}_{t+12}))$                                             |                      | 3.892**<br>(1.794)   | 3.552*<br>(1.917)    | 3.445*<br>(1.917)    | 3.817**<br>(1.772)   |
| $E_t(\Delta \Pr(U_{t+12} \geq 0)) \times$<br>$E_t(\Pr(\text{Find Job}_{t+3}))$                                              |                      |                      | -0.224<br>(1.335)    | 0.036<br>(1.340)     | 0.909<br>(1.259)     |
| $E_t(\Pr(\text{Lose Job}_{t+12})) \times$<br>$E_t(\Pr(\text{Find Job}_{t+3}))$                                              |                      |                      |                      | -1.364<br>(1.456)    | -0.794<br>(1.442)    |
| $E_t(\Delta \Pr(U_{t+12} \geq 0)) \times$<br>$E_t(\Pr(\text{Lose Job}_{t+12})) \times$<br>$E_t(\Pr(\text{Find Job}_{t+3}))$ |                      |                      |                      |                      | -13.438**<br>(5.442) |
| $E_t(\pi_{t+12})$                                                                                                           | -0.023<br>(0.023)    | -0.024<br>(0.023)    | -0.023<br>(0.023)    | -0.024<br>(0.023)    | -0.023<br>(0.023)    |
| F-stat                                                                                                                      |                      | 4.71                 |                      |                      | 6.10                 |
| p-value                                                                                                                     |                      | 0.030                |                      |                      | 0.014                |
| Observations                                                                                                                | 4130                 | 4130                 | 4130                 | 4130                 | 4130                 |
| Adj. R-squared                                                                                                              | 0.177                | 0.179                | 0.179                | 0.179                | 0.181                |

Notes: Standard errors in parentheses. Dependent variable is weekly job search hours (win-sorized). All specifications include state×time fixed effects and full controls. Standard errors are clustered by individual. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

expectations are included. Macro beliefs primarily shift intensity among those already searching, while personal job-loss beliefs operate along both margins.

*Timing and reverse causality.* To address concerns that search may influence expectations rather than the reverse, Table A7 uses lagged expectations from the prior core survey wave, which precedes the LMM by at least one month. The lagged job-loss expectations coefficient remains positive and, if anything, stronger than the contemporaneous estimate. This timing is consistent with expectations preceding search.

*Within-person variation.* First-difference and individual fixed effects specifications exploit within-worker variation over time (Table A8). The job-loss coefficient remains positive but becomes imprecise, reflecting the short panel dimension: workers appear in the LMM only about 1.4 times on average. Voluntary-separation beliefs remain predictive in some within-person specifications, but this margin is ancillary to the precautionary interpretation emphasized here.

*Employed versus unemployed workers.* Table A9 compares the role of expectations by em-



ployment status. Among employed workers, macro pessimism predicts search intensity. Among the unemployed, the relationship is essentially zero. This asymmetry supports the paper’s focus on the employed margin: expectations matter for workers who have a job to insure, not for those already unemployed and searching by necessity.

## 2.7 External validation in the ECB Consumer Expectations Survey

The SCE is well suited to the main analysis because it combines aggregate expectations, personal labor-market expectations, detailed search hours, stated search motives, and panel outcomes for the same employed workers. The ECB Consumer Expectations Survey provides a useful external check on the central mechanism in a different setting.<sup>8</sup>

Appendix Table A13 reports three exercises using employed respondents in the ECB labor-market module matched to monthly expectations. The first two reproduce the main links behind the SCE mechanism. First, macroeconomic pessimism predicts higher perceived job-loss risk. After controlling for other macro and household expectations, a 10 percentage point higher expected unemployment rate is associated with about a 0.9 percentage point higher perceived probability of losing the current job over the next 3 months. Second, perceived job-loss risk predicts search. A 10 percentage point higher perceived job-loss probability is associated with a 4.7 percentage point higher probability of actively looking for a job and a 6.2 percentage point higher reported probability of looking for a job over the next 3 months. Search intensity also rises with perceived job-loss risk.

The ECB evidence therefore supports the external validity of the paper’s core mechanism: workers who expect worse aggregate labor-market conditions perceive higher own job-loss risk, and workers who perceive higher job-loss risk search more. The ECB is less suited to the richer heterogeneity tests in the SCE. In the regular quarterly module, the core job-finding probability is observed for unemployed respondents, not employed respondents. The topical labor-market module contains an employed-worker analogue, expected time to find an equivalent job, but it is measured in a separate module rather than in the same quarterly interview. I therefore use it only as a supplementary predetermined-belief check, merging the round-41 topical response forward to later quarterly observations and,

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<sup>8</sup>I use the ECB CES microdata release dated 28 April 2026, covering the regular monthly and quarterly modules through 2026Q1. The regular quarterly module measures job-loss risk over a 3-month horizon and contains search outcomes such as active search, the probability of looking for a job, and search intensity. The analysis uses the 11 euro area countries in the release. All specifications include survey-round and country fixed effects, demographic controls, ECB CES weights, and respondent-clustered standard errors. The controlled specifications additionally include expected inflation, expected economic growth, current perceived unemployment, and expected household financial conditions. Expected unemployment is also included as a control in the search regressions, where it is not the focal regressor.

more tightly, to the nearest subsequent quarterly round. This exercise does not replicate the SCE heterogeneity result: in the ECB data, the search response to job-loss risk is not stronger among workers who expect it would take longer to find an equivalent job. I therefore use the ECB as evidence that the belief-to-risk-to-search chain is not unique to the SCE, while the direct evidence on precautionary motives and heterogeneity by job-finding beliefs comes from the SCE.

The estimates in this section are associations, not causal effects. Identifying the causal effect of expectations on search would require variation that moves what workers believe about the labor market without moving their search incentives through some other channel, and such variation is hard to come by: the aggregate shocks that shift beliefs also shift labor-market conditions, and the personal shocks that shift beliefs often shift search directly. What the cross-sectional design delivers instead is a comparison of workers in the same time and place who happen to disagree about the economy. That comparison cannot rule out unobserved worker or job heterogeneity, but three features of the data make a pure omitted-variables story uncomfortable: the beliefs predict realized separations, they predict security-related search motives, and they matter most exactly where an outside offer is most valuable.

It is still worth asking what makes workers in the same state and month disagree about the macroeconomy in the first place. The 2016 presidential election offers one clean illustration. Following [Coibion et al. \(2020\)](#) and [Mian et al. \(2018\)](#), I compare workers in Republican- and Democrat-won states (dropping swing states) around November 2016: partisan priors interacting with the election outcome opened up a roughly 6 percentage point gap in macro unemployment expectations across otherwise similar local labor markets, with flat pre-trends, and the gap did *not* carry over to personal perceived job-loss risk—workers separate their views about the aggregate economy from the read on their own job security, the same asymmetry documented in [Section 2.3](#). The implied search response has the expected sign but is imprecise at the quarterly frequency of the module. I read the episode as an illustration of where within-cell belief dispersion comes from rather than as a stand-alone identification strategy, and report the full event-study and difference-in-differences estimates in [Appendix B.6](#).

Taken together, the results show that expectations about the aggregate economy are tied to how hard employed workers search, and that the link runs mainly through perceived job-loss risk. The aggregate-pessimism coefficient survives controlling for personal beliefs, which suggests that it still captures perceived risk beyond what the SCE’s personal-expectation questions span. The model in the next section formalizes this expectations-search link and traces its aggregate implications.

### 3 Model

The empirical analysis documented three facts about employed workers in the Survey of Consumer Expectations. Workers who report higher job-loss risk search more while employed. This response is stronger when expected re-employment prospects are weak and is reflected in job-security search motives. Reported job-loss probabilities also predict subsequent separations.

This section develops a Diamond–Mortensen–Pissarides search-and-matching model with on-the-job search. The model has two main ingredients: separation risk rises in recessions, and employed workers choose search effort before a possible job loss. When a separation occurs, a worker with an outside offer moves directly into a new job next month, while a worker without an offer enters unemployment. This is the mechanism emphasized by [Ahn and Shao \(2017\)](#): on-the-job search is valuable not only because it may improve the current job ladder, but also because it creates an option that is useful when the worker faces job loss. Search therefore affects unemployment dynamics by changing the fraction of separations that become unemployment spells.

The model abstracts from voluntary job-to-job transitions absent separation, heterogeneous match quality, and endogenous separations. These simplifications keep the focus on the response of employed search to perceived job-loss risk. I also add a job-ladder extension. The extension uses the same offer technology, but allows outside options to have two uses. If the current job survives, an offer can generate an ordinary job-ladder move. If the current job is destroyed, the outside option can insure the separation. Matching total E-to-E mobility and the share of E-to-E moves related to job loss pins down how much of the offer portfolio is usable at separation. The baseline shows the mechanism in its cleanest form. The extension asks how much of it remains once ordinary job-ladder mobility is also matched.

Section 4 describes the calibration, and Section 5 uses the calibrated model to quantify the aggregate implications of precautionary on-the-job search.

#### 3.1 Environment

Time is discrete and indexed by  $t = 0, 1, 2, \dots$ , with each period interpreted as one month. The economy is populated by a unit mass of risk-neutral workers and a continuum of risk-neutral firms. Workers discount future payoffs at factor  $\beta \in (0, 1)$ . Firms post vacancies at flow cost  $\kappa > 0$  and enter freely.

**Aggregate states and separations.** The aggregate state is  $z \in \{G, B\}$ , where  $G$  denotes an expansion and  $B$  a recession. Productivity is  $y(z)$ , and the state follows a Markov transition matrix  $\Pi(z, z')$ . An employed worker's current match is destroyed during the month with probability  $\delta(z)$ . Separation risk is higher in recessions,  $\delta(B) > \delta(G)$ .

**Beliefs.** Workers may perceive separation risk differently from the objective probability. Let  $\tilde{\delta}(z)$  denote the worker's perceived monthly probability of job loss. In the calibration, a level parameter  $\zeta$  matches the average level of perceived job-loss risk and a state component  $m(z)$  allows beliefs to move differently across aggregate states.<sup>9</sup> Beliefs affect search choices. Realized worker flows are governed by the objective separation probability  $\delta(z)$ .

**Timing.** Within a period  $t$ , the realized aggregate state is  $z_t$ . Firms post vacancies and matching occurs. Employed workers choose search intensity  $s(z_t)$  and may receive outside offers. Separations are realized at the end of the month. Matches formed during month  $t$  become active at  $t + 1$  and are therefore not exposed to an additional separation draw within their month of formation. The economy then transitions to  $z_{t+1}$  according to  $\Pi$ , and continuation values are evaluated in  $z_{t+1}$ .<sup>10</sup> This timing is convenient empirically: it matches the monthly frequency of the SCE and the notion that current labor-market conditions govern offers and separations over the month, while expectations about the future state matter through discounted continuation values.

**Matching technology** Let  $u$  denote unemployment,  $v$  vacancies, and  $e = 1 - u$  employment at the beginning of the month. Unemployed workers supply one unit of search. Employed workers supply effective search units  $g(s(z))$ , where  $g(s) = \log(1 + s)$ . Total search units are

$$S(z) = u(z) + e(z)g(s(z)).$$

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<sup>9</sup>The exogenous separation block is a reduced-form representation of the job-loss risk workers report in the survey, not a theory of why matches end. In a richer model, match-specific productivity, firm distress, or worker-firm sorting would generate heterogeneous and partly endogenous separation probabilities. Those forces would change the source of  $\delta(z)$ , but the employed worker's search problem would still depend on the perceived probability of job loss and the value of having an outside offer when that loss occurs. I therefore use  $\tilde{\delta}(z)$  as the state variable pinned down by the SCE beliefs and keep the destruction process outside the model. This choice isolates the precautionary-search margin, while leaving endogenous quits, match-quality selection, and the effect of search on match destruction for richer extensions.

<sup>10</sup>Thus, within-month transition probabilities (job finding, offer arrival, separations) are functions of the *current* state  $z_t$ , while  $\Pi$  governs only the distribution of next month's continuation values.

Following the searcher-pool formulation in [Mukoyama et al. \(2018\)](#), market tightness is vacancies per search unit,

$$\theta(z) = \frac{v(z)}{S(z)}.$$

The matching function is

$$M(S, v) = \chi S^\alpha v^{1-\alpha}, \quad \alpha \in (0, 1), \chi > 0.$$

Tightness summarizes congestion in the pool of searchers: when  $\theta$  rises, searchers are more likely to meet firms, while vacancies are harder to fill. The monthly job-finding probability for an unemployed worker in state  $z$  is

$$p_u(z) = 1 - \exp[-\chi \theta(z)^{1-\alpha}], \quad (2)$$

and the probability a vacancy is filled in state  $z$  is

$$p_f(z) = 1 - \exp[-\chi \theta(z)^{-\alpha}]. \quad (3)$$

### 3.2 Employed Search and Offers

An employed worker chooses search intensity  $s \geq 0$  at flow cost

$$c(s) = \frac{\gamma}{2} s^2.$$

Search produces an outside offer with probability

$$p_{\text{offer}}(z, s) = 1 - \exp\{-\eta \chi \theta(z)^{1-\alpha} \psi g(s)\}, \quad g(s) = \log(1 + s).$$

Its derivative with respect to search effort is

$$p_{\text{offer},s}(z, s) = \exp\{-\eta \chi \theta(z)^{1-\alpha} \psi g(s)\} \eta \chi \theta(z)^{1-\alpha} \psi g'(s). \quad (4)$$

The parameter  $\psi$  governs the effectiveness of employed search effort and  $\gamma$  governs the cost of search. The parameter  $\eta$  allows employed search to be less effective than unemployed search per unit of effort.

The derivative  $p_{\text{offer},s}(z, s)$  is the marginal return to employed search effort.<sup>11</sup> The

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<sup>11</sup>[Graves and Huckfeldt \(2026\)](#) make a related distinction between active search, in which workers deliberately apply or contact employers, and passive search, in which jobs arrive through referrals or other unsolicited channels. Their evidence is useful for separating search effort from the return to that effort.

precautionary motive does not require this return to rise in recessions. Beliefs matter by raising the value of search as insurance against job loss.

The same offer technology can serve two roles. If the current job survives, an offer may generate an ordinary job-ladder move. If the current job is destroyed, an offer may allow the worker to avoid unemployment. Let  $\rho_J$  be the probability that an offer is acceptable as a voluntary job-ladder move conditional on no separation. Let  $\rho_I$  be the probability that an offer is usable as insurance conditional on separation. The first parameter pins down ordinary E-to-E mobility; the second pins down how much of the offer portfolio protects workers at separation. The baseline outside-option model is the special case  $\rho_J = 0$  and  $\rho_I = 1$ .

The model uses a parsimonious reduced-form value for ladder moves. In the main specification, an acceptable ladder offer gives a gain proportional to the employment surplus:

$$\Delta_J(z') = \omega_J(W(z') - U(z')).$$

This keeps the ladder value tied to the same Bellman objects that determine the insurance value of avoiding unemployment. The parameter  $\omega_J$  is calibrated from the average wage gain of E-to-E movers.

### 3.3 Worker Values

Let  $W(z)$  and  $U(z)$  denote the continuation values of employment and unemployment, evaluated at the worker's search policy and under the objective law of motion for separations. To keep the notation light, write  $p_e(z) \equiv p_{\text{offer}}(z, s(z))$ . The value of employment averages over two events. If the match survives, a usable offer may generate a ladder gain. If the match is destroyed, a usable offer keeps the worker employed next period; otherwise the worker enters unemployment:

$$\begin{aligned} W(z) = w(z) - c(s(z)) + \beta \sum_{z'} \Pi(z, z') \{ & (1 - \delta(z)) [W(z') + \rho_J p_e(z) \Delta_J(z')] \\ & + \delta(z) [\rho_I p_e(z) W(z') + (1 - \rho_I p_e(z)) U(z')] \}. \end{aligned} \quad (5)$$

The unemployed value is

$$U(z) = b + \beta \sum_{z'} \Pi(z, z') [p_u(z) W(z') + (1 - p_u(z)) U(z')].$$

When choosing search, workers place perceived risk on the separation branch. The first-order condition for employed search is

$$c'(s(z)) = \beta p_{\text{offer},s}(z, s(z)) \sum_{z'} \Pi(z, z') [(1 - \tilde{\delta}(z)) \rho_J \Delta_J(z') + \tilde{\delta}(z) \rho_I (W(z') - U(z'))]. \quad (6)$$

The bracketed term separates the two motives for search. The first term is the ladder motive: an offer is valuable because it may improve the worker's job if the current match survives. The second is the precautionary motive: an offer is valuable because it may prevent unemployment if the current match is destroyed. A higher perceived separation probability raises the value of search as insurance against job loss. Search rises with perceived risk when this insurance value is large enough relative to the ladder value displaced by the higher chance of separation.

### 3.4 Flows

Realized flows are governed by objective separation risk. The employment-to-unemployment probability is

$$p_{EU}(z) = \delta(z) [1 - \rho_I p_{\text{offer}}(z, s(z))]. \quad (7)$$

Search lowers unemployment inflows only through the insurance component. Voluntary job-ladder moves change jobs but do not directly prevent unemployment.

The model generates two E-to-E flows:

$$p_{EE}^J(z) = (1 - \delta(z)) \rho_J p_{\text{offer}}(z, s(z)),$$

and

$$p_{EE}^I(z) = \delta(z) \rho_I p_{\text{offer}}(z, s(z)).$$

Total E-to-E mobility is  $p_{EE}^J(z) + p_{EE}^I(z)$ . The ratio

$$\frac{p_{EE}^I(z)}{p_{EE}^J(z) + p_{EE}^I(z)}$$

maps the model to the observed share of E-to-E moves that are associated with job loss or anticipated layoff. This moment is informative about  $\rho_I$ : self-reported job-loss E-to-E transitions are a conservative measure of insured separations, because workers who anticipate job loss and move successfully may describe the move as a better-job transition rather than a layoff.



### 3.5 Firm Values and Equilibrium

Ladder moves generate quit risk for firms. A filled job survives to the next month only if the match is not destroyed and the worker does not leave for an acceptable ladder offer. The firm value is

$$J(z) = y(z) - w(z) + \beta(1 - \delta(z)) \sum_{z'} \Pi(z, z') [(1 - \rho_J p_{\text{offer}}(z, s(z)))J(z')].$$

The insurance parameter  $\rho_I$  does not enter firm survival because an insured separation uses an offer only after the current match has already been destroyed. Free entry sets the cost of a vacancy equal to the expected value of a filled job.

Wages are determined by Nash bargaining over the full equilibrium match values, not by the closed-form DMP wage that would apply without on-the-job search. Let the worker's match surplus be

$$X(z) = W(z) - U(z).$$

Taking the search policy, offer probability, and quit risk as part of the equilibrium, the wage in state  $z$  solves

$$\max_{w(z)} X(z)^\phi J(z)^{1-\phi}.$$

Since the wage is a transfer between the worker and firm, the Nash condition is

$$(1 - \phi)X(z) = \phi J(z). \tag{8}$$

Perceived separation risk affects bargaining only through the search policy. Wages are bargained over objective match values, rather than over each worker's subjective belief about job loss. One interpretation is a representative bargaining protocol in which the wage uses the common law of motion for jobs and firms, while individual workers search more when they perceive higher own job-loss risk. This restriction shuts down the belief-to-wage channel studied by [Menzio \(2022\)](#) and leaves the precautionary-search channel as the only direct role for beliefs.

Equation (8) is solved jointly with the worker value, firm value, free-entry, and search equations. In the job-ladder extension,  $X(z)$  includes the worker's ladder and insurance value, while  $J(z)$  includes the quit-survival term  $(1 - \rho_J p_{\text{offer}}(z, s(z)))$ . An equilibrium consists of worker values, firm values, wages, tightness, employed search, and flows such that employed search satisfies (6), firms satisfy free entry, wages satisfy (8), and stocks obey the law of motion implied by  $p_{EU}$  and  $p_u$ .

### 3.6 Analytical Implications

Equation (6) gives the main mechanism. Search has a ladder value and an insurance value. Higher perceived job-loss risk shifts weight toward the insurance branch. Whether search rises depends on which branch is more valuable at the margin.

**Assumption 1** (Regularity). *The employed-search choice is interior,  $p_{offer,s}(z, s) > 0$ , and  $c''(s) > 0$ .*

**Proposition 1** (Perceived risk raises search when the insurance value is high). *Holding tightness, continuation values, and ladder values fixed, employed search is increasing in perceived separation risk whenever*

$$\sum_{z'} \Pi(z, z') \rho_I (W(z') - U(z')) > \sum_{z'} \Pi(z, z') \rho_J \Delta_J(z').$$

This is the model counterpart of the empirical mechanism. When workers think job loss is more likely, an offer is less about improving a surviving match and more about avoiding unemployment. If that insurance value is large enough, perceived risk raises search.

**Proposition 2** (Search lowers unemployment inflows when offers insure separations). *If  $\rho_I > 0$  and  $p_{offer,s}(z, s) > 0$ , then  $p_{EU}(z)$  is decreasing in employed search effort.*

The sign is immediate from (7). More search raises the probability of having an offer in hand; if some offers can be used at separation, fewer separations become unemployment spells.

**Proposition 3** (A sufficient condition for countercyclical search). *Search is higher in recessions than expansions if the recession-state marginal benefit of search in (6) is at least as large as the expansion-state marginal benefit at the relevant effort levels.*

This condition makes clear why countercyclical search is not automatic. Recessions lower tightness, which makes effort less effective. Search rises only if the higher perceived-risk motive is strong enough to offset that force and any changes in surplus or ladder value. This is exactly what the counterfactuals below test: when perceived job-loss probabilities are held fixed across states, the precautionary motive loses its cyclical force and search no longer rises in the same way.

**Proposition 4** (Precautionary search dampens recession unemployment). *For a given recession equilibrium, if endogenous search exceeds the fixed-search level and  $\rho_I > 0$ , then unemployment is weakly lower than in the fixed-search counterfactual.*

The proposition pins down the sign, not the size. The size depends on how many offers are actually usable when a job ends. That is why the calibration separates ordinary ladder mobility from insurance against job loss. Once that distinction is imposed, beliefs still matter for worker behavior, but the aggregate unemployment effect is disciplined by the share of offers that can prevent unemployment.

## 4 Calibration

The calibration uses two kinds of evidence. Aggregate stock-flow moments pin down the unemployment accounting. SCE moments pin down beliefs, offer receipt, and the response of search to perceived job-loss risk. I first report the baseline outside-option model. I then report the job-ladder extension, which adds E-to-E mobility and the job-loss-related share of those moves as targets. The extension asks how much of the baseline mechanism remains once ordinary job-ladder mobility is also matched.

### 4.1 Targets

Time is monthly. I set standard parameters outside the calibration:  $\beta = 0.9967$ ,  $b = 0.4$ ,  $\alpha = 0.5$ ,  $\phi = 0.5$ , and an employed/unemployed contact ratio  $\eta = 0.10$ . Productivity takes two values,  $y(G) = 1$  and  $y(B) = 0.95$ , with state persistence chosen to match NBER business-cycle durations. Appendix Table A15 reports the full list and sources.

The main calibration uses the SCE profile. It matches mean perceived 12-month job-loss risk in the search sample, 0.212; the employed offer probability, 0.330; and the empirical semi-elasticity of search with respect to perceived job-loss risk,  $\beta_{\text{plse}} = 1.688$ . In the job-ladder extension, it also matches total monthly E-to-E mobility, 0.0385, and the share of E-to-E moves reported as job-loss or anticipated-layoff moves, 0.094.

These last two moments separate ordinary ladder moves from offers that insure job loss. Total E-to-E mobility identifies  $\rho_J$ , the probability that an offer is acceptable as a ladder move. The job-loss-related E-to-E share identifies  $\rho_I$ , the probability that an offer is usable at separation. I treat the latter moment as conservative. A worker who sees a layoff coming, searches, and moves directly to a better job may report the move as a better-job transition rather than as job loss. For this reason, I also report a calibration with higher insurance use,  $\rho_I = 0.75$ .

The ladder value parameter  $\omega_J$  is calibrated from the average wage gain of E-to-E movers. In the headline specification, the gain is proportional to the employment surplus,  $\Delta_J = \omega_J(W - U)$ . Thus  $\omega_J$  is the ladder-value parameter;  $\Delta_J$  is not estimated as a separate

level.

**Stock-flow discipline.** The unemployment stock and the U-to-E probability cannot be combined with the raw SCE E-to-U transition as three independent targets. In a two-state steady state,  $u = 0.0573$  and  $p_{UE} \simeq 0.28$  imply

$$p_{EU} = \frac{up_{UE}}{1-u} \simeq 0.0170. \quad (9)$$

The raw SCE next-wave E-to-U probability is lower, about 0.0116, reflecting time aggregation, survey rotation, and movements through inactivity. I therefore target the unemployment stock and the U-to-E probability, and report the implied E-to-U probability as the stock-flow object. The objective job-loss probability is recalibrated after each update of  $\rho_I$ , because the effective E-to-U probability is

$$p_{EU}(z) = \delta(z) [1 - \rho_I p_{\text{offer}}(z, s(z))].$$

This ordering matters. Lowering  $\rho_I$  reduces the fraction of separations covered by offers. To keep the stock-flow E-to-U probability fixed, the underlying job-loss probability must then rise.

## 4.2 Calibration Results

The baseline outside-option calibration sets  $\rho_I = 1$  and  $\rho_J = 0$  by construction. It matches the same perceived-risk, offer-probability, and search-response anchors. The calibrated belief shifter is  $\zeta = 0.775$ , search cost curvature is  $\gamma = 0.0248$ , and search effectiveness is  $\psi = 27.11$ . In equilibrium, search rises from 0.537 in expansions to 0.678 in recessions, while the employed offer probability rises from 0.330 to 0.363.

Table 7 reports the targeted moments and calibrated parameters for the job-ladder extension. The model matches the micro anchors: mean perceived job-loss risk is 0.212, the employed offer probability is 0.330, and the model-implied search semi-elasticity is 1.688. It also matches total E-to-E mobility and the job-loss-related E-to-E share. Appendix Table A16 reports untargeted cyclical moments.

**Interpretation.** The calibrated  $\rho_I = 0.532$  implies that slightly more than half of offer opportunities can serve as insurance against job loss. I do not read this as the literal share of workers who receive clean advance notice of a layoff; it is the model's mapping from reported job-loss E-to-E moves into the share of offers that prevent separations from

**Table 7.** Calibration Targets and Parameter Values: Job-Ladder Extension

| <b>Panel A. Calibration Targets</b>            |                                 |          |                             |
|------------------------------------------------|---------------------------------|----------|-----------------------------|
| Moment                                         | Data                            | Model    | Role                        |
| Unemployment rate $u$                          | 0.057                           | 0.054    | stock-flow target           |
| Monthly U-to-E probability $p_{UE}$            | 0.280                           | 0.280    | stock-flow target           |
| Stock-flow E-to-U probability $p_{EU}$         | 0.017                           | 0.017    | implied by $u$ and $p_{UE}$ |
| Mean perceived job-loss probability, 12 months | 0.212                           | 0.212    | target                      |
| Employed offer probability                     | 0.330                           | 0.330    | target                      |
| Search semi-elasticity $\beta_{\text{plse}}$   | 1.688                           | 1.688    | target                      |
| Monthly total E-to-E probability               | 0.0385                          | 0.0383   | target                      |
| Job-loss-related share of E-to-E               | 0.094                           | 0.095    | target                      |
| <b>Panel B. Calibrated Parameters</b>          |                                 |          |                             |
| Parameter                                      | Description                     | Value    |                             |
| $\chi$                                         | Matching efficiency             | 0.332    |                             |
| $\kappa$                                       | Vacancy cost                    | 0.561    |                             |
| $\psi$                                         | Search effectiveness            | 8.696    |                             |
| $\gamma$                                       | Search cost curvature           | 0.000674 |                             |
| $\zeta$                                        | Belief level shifter            | 0.956    |                             |
| $\rho_I$                                       | Offer usable as insurance       | 0.532    |                             |
| $\rho_J$                                       | Offer acceptable as ladder move | 0.108    |                             |
| $\omega_J$                                     | Ladder gain as share of surplus | 0.145    |                             |

*Notes:* The model uses the proportional-surplus ladder specification,  $\Delta_J = \omega_J(W - U)$ . The stock-flow E-to-U probability is implied by the unemployment and U-to-E targets rather than independently targeted from raw SCE transitions.

becoming unemployment spells. The calibrated  $\rho_f = 0.108$  lets the model match total E-to-E flows without treating all mobility as precautionary insurance. Once this ladder margin is included, most employed search is no longer unemployment insurance. The precautionary component still matters for worker behavior, because it makes search respond to perceived job-loss risk, but it is not the dominant source of E-to-E mobility. Appendix Tables [A17](#) and [A21](#) report robustness profiles and sensitivity checks.

## 5 Quantitative Results

The model has two primary results. First, it studies the mechanism linking aggregate beliefs to employed search. Pessimistic beliefs raise search because they make an outside offer more valuable as insurance against job loss, as in Proposition 1. Second, this precautionary motive helps explain why employed search can rise in recessions, even when vacancies are scarce and offers are harder to get. The aggregate unemployment effect is more limited once the model also matches ordinary job-to-job mobility. Beliefs matter for worker behavior, but whether they matter for unemployment depends on whether outside offers can actually be used when a separation occurs.

### 5.1 Search Cyclicalities

Table 8 reports search and offer probabilities in expansions and recessions, the quantitative counterpart of Proposition 3. In the baseline model, search rises from 0.6 in expansions to 0.7 in recessions, an increase of 26 percent. This is not because the calibration targets search separately in the two states. The calibration targets the worker-level response of search to perceived job-loss risk. In a recession, workers search more because the insurance value of an outside offer rises.

This goes against the usual tightness argument. In recessions, vacancies are scarce relative to searchers, so offers are harder to get. If that were the only force, employed search would fall. In the calibrated model, higher perceived job-loss risk leads to the precautionary motive to dominate and search rises.<sup>12</sup>

Here the key takeaway is that beliefs can affect search response so as to make a recession more search-intensive. A recession does not only lower the return to effort by reducing tightness; it also raises the value of avoiding unemployment. The model shows that the

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<sup>12</sup>Search effort and offer probabilities need not move one-for-one, because tightness in recession worsens. In the baseline model, higher recession search raises the offer probability from 0.330 to 0.363. In the job-ladder extension, lower tightness offsets more of the increase in effort, so the offer probability is roughly unchanged.

second force can dominate. Expectations about aggregate economy can help generate countercyclical search: they make workers search harder precisely when outside offers are harder to obtain.

The lower rows of Table 8 show the same comparison in the job-ladder extension. The extension is useful because it distinguishes insurance from job-to-job mobility. Not every E-to-E move is insurance against unemployment. Many are ordinary ladder moves. The extension separates these two uses of outside offers. Even after doing so, search is higher in the recession state, but the offer probability is roughly flat because lower tightness offsets much of the extra effort.<sup>13</sup>

**Table 8.** Search and Offer Probabilities Across States

| Profile                               | $s(G)$ | $s(B)$ | Change in $s$ | $p_{\text{offer}}(G)$ | $p_{\text{offer}}(B)$ |
|---------------------------------------|--------|--------|---------------|-----------------------|-----------------------|
| Baseline outside-option model         | 0.537  | 0.678  | 26.3%         | 0.330                 | 0.363                 |
| Job-ladder extension, SCE core        | 3.022  | 3.214  | 6.4%          | 0.330                 | 0.322                 |
| Higher insurance use, $\rho_I = 0.75$ | 2.089  | 2.282  | 9.2%          | 0.330                 | 0.327                 |
| CPS                                   | 1.933  | 2.127  | 10.0%         | 0.330                 | 0.329                 |

*Notes:* Each row is recalibrated, except that the higher-insurance-use case fixes  $\rho_I = 0.75$ . Search is in model effort units. The calibration targets the response of search to perceived job-loss risk, not the level of search in each state.

## 5.2 Aggregate Recession Counterfactual

Table 9 reports the main recession counterfactual, which quantifies the dampening result in Proposition 4. It asks how much aggregate unemployment would increase if workers did not raise search in recessions. I keep the economy in the recession state, but force employed search to stay at its expansion level. I then compare recession unemployment under fixed search with recession unemployment when workers choose search optimally.

The first row of Table 9 shows the baseline outside-option model. Unemployment rises from 5.24 percent in expansions to 8.42 percent in recessions. If employed search is held fixed at its expansion level, recession unemployment rises to 9.03 percent. The difference is 0.61 percentage points, or about 16 percent of the fixed-search recessionary increase. In this benchmark, precautionary search is an economically meaningful stabilizer: workers' extra search offsets roughly one sixth of the recessionary rise in unemployment.

The remaining rows of Table 9 show how the job-ladder extension changes the conclu-

<sup>13</sup>The robustness rows show that the search result is not driven by one aggressive reading of E-to-E moves. When a larger share of offers can be used at separation, search still rises in recessions. In the CPS profile, less E-to-E mobility is assigned to the insurance channel, but search also remains higher in the recession state. The behavioral result is therefore stable: when perceived job-loss risk rises, workers search more.

**Table 9.** Recession Steady States: Baseline vs. Fixed Search

| Profile                        | $\rho_I$ | $\rho_J$ | $u(G)$ | $u(B)$ | $u(B)$ fixed $s$ | Dampening        |
|--------------------------------|----------|----------|--------|--------|------------------|------------------|
| Baseline outside-option model  | 1.000    | 0.000    | 5.24%  | 8.42%  | 9.03%            | 0.61 pp (16.0%)  |
| Job-ladder extension, SCE core | 0.532    | 0.108    | 5.39%  | 9.11%  | 9.15%            | 0.046 pp (1.22%) |
| Higher insurance use           | 0.750    | 0.102    | 5.39%  | 9.10%  | 9.21%            | 0.112 pp (2.93%) |
| CPS                            | 0.223    | 0.079    | 5.40%  | 9.06%  | 9.09%            | 0.030 pp (0.81%) |

Notes: Dampening is  $100 \times [u_B^{\text{fixed search}} - u_B^{\text{reoptimized}}]$ . The fixed-search path holds recession employed search at the expansion equilibrium level while using recession-state contact probabilities.

sion. In the SCE core calibration, precautionary search lowers recession unemployment by 0.046 percentage points. When more offers can be used at separation, the effect rises to 0.112 percentage points. This is the force in Proposition 2: search lowers unemployment inflows only through offers that are usable at separation. Once job-to-job moves are matched separately, only a part of the offer portfolio is available as insurance against job loss. Search still responds to beliefs, but only a small share of the extra offers prevents unemployment.

The baseline model shows that the mechanism can matter for aggregates if outside offers mainly insure job loss. The job-ladder extension then disciplines that channel using E-to-E mobility. It says that much of employed search is ordinary mobility rather than unemployment insurance. The behavioral result survives, but the aggregate unemployment effect becomes modest. Beliefs matter a lot for what workers do; they matter for unemployment only to the extent that the additional search produces offers that are actually usable at separation.

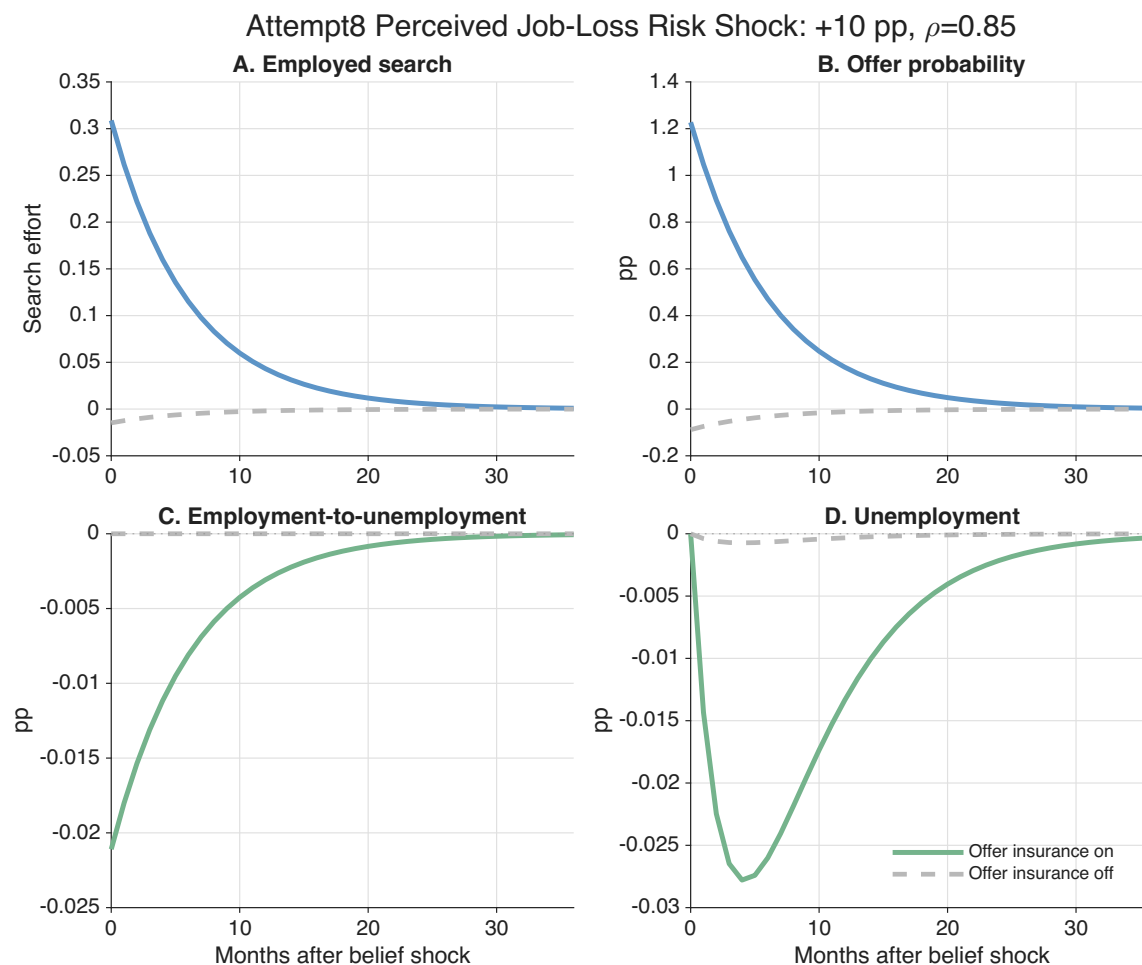
Figure 3 reports the belief-shock exercise, which isolates the search decision from changes in actual labor-market conditions. This exercise is the cleanest quantitative version of Proposition 1. The aggregate state, objective separation risk, productivity, and ladder parameters are held fixed. I then raise perceived 12-month job-loss risk by 10 percentage points and let the belief shock fade with monthly persistence  $\rho = 0.85$ .

The first panels of Figure 3 show the behavioral response. Workers search more because an outside offer is more valuable when they think job loss is more likely. At impact, perceived 12-month job-loss risk rises from 32 percent to 42 percent, search rises by about 7 percent, and the offer probability rises by 1.5 percentage points. This isolates the belief mechanism from recession fundamentals: even with actual separation risk and productivity fixed, perceived risk alone raises search.

The E-to-U probability falls by only 0.009 percentage points, because only some offers can be used at separation. This small aggregate response is not a failure of the belief mechanism. It is a statement about where the mechanism operates. Beliefs first move



worker behavior. They affect unemployment only through the narrower channel of offers that arrive before, and can be used at, job loss.



*Notes:* The figure reports responses in the calibrated job-ladder extension to a 10 percentage point increase in perceived 12-month job-loss risk with monthly persistence  $\rho = 0.85$ . Actual labor-market conditions are held fixed. The solid line keeps the offer-insurance channel active. The dashed line sets the insurance use of offers to zero. Outcomes are deviations from the no-shock path.

**Figure 3.** Impulse Response to a Perceived Job-Loss Risk Shock

The two lines in Figure 3 show what search can and cannot do. When offers can insure separations, worse beliefs raise search and slightly reduce E-to-U transitions. When offers cannot be used at separation, search still rises, but unemployment does not fall. Beliefs move search through the value of an outside offer. Unemployment moves only if that offer can actually keep a separated worker employed.

### 5.3 Discussion

The counterfactuals put the empirical results in perspective. The survey evidence shows a strong behavioral margin: workers who perceive higher job-loss risk search more while still employed. The model helps disentangle the mechanism behind this relationship. Beliefs affect search through a precautionary motive—when workers think job loss is more likely, an outside offer becomes more valuable because it can keep a separation from becoming an unemployment spell.

This mechanism also helps explain countercyclical search. In recessions, lower tightness makes offers harder to get, which by itself would discourage search. At the same time, perceived job-loss risk is higher, so the insurance value of an offer rises. When this precautionary force is strong enough, employed search rises even though the market is less favorable.

However, the same mechanism does not automatically imply a large aggregate unemployment effect. Search affects unemployment only through the offers that are available at the moment of separation. If an extra offer would have produced an ordinary job-to-job move anyway, it changes the employer a worker moves to but does not prevent an unemployment spell. If the offer is available when a job is lost, it can turn a potential E-to-U transition into an E-to-E transition. The unemployment effect therefore depends on the composition of outside offers, not only on how much search rises.

In the baseline model, most offers serve the insurance role by construction, so precautionary search offsets a meaningful share of the recessionary rise in unemployment. Once the model also matches ordinary E-to-E mobility, many offers are instead assigned to ladder mobility. Beliefs still move search, and the model generates countercyclical employed search, but the aggregate effect becomes smaller because only part of the additional search is useful as insurance against job loss.

The main takeaway is therefore that macro as well as personal beliefs are important for workers' choices, and they help explain why employed search rises when the economy weakens. However, beliefs matter for aggregate unemployment only to the extent that the search they induce produces offers that can be used at separation. The model's contribution is to make that distinction explicit by separating search for better jobs from search that insures job loss.

## 6 Conclusion

This paper studies how beliefs about the economy affect job search while workers are still employed. The empirical results point to a simple channel. Workers who are more pessimistic about the macroeconomy perceive a higher risk of losing their own job, and workers who perceive higher job-loss risk search more. In the Survey of Consumer Expectations, perceived job-loss risk predicts search intensity, forecasts subsequent separations, and predicts job-security motives among workers who search. The response is also strongest when workers see both high job-loss risk and weak re-employment prospects, exactly the case in which having an outside offer is most valuable. The ECB Consumer Expectations Survey shows the same first-order pattern in a different setting: aggregate pessimism predicts perceived job-loss risk, and perceived job-loss risk predicts search.

I interpret these facts as evidence of precautionary on-the-job search. Employed workers search not only to move up the job ladder, but also to reduce the chance that job loss turns into unemployment. The model formalizes this distinction. Actual job-loss risk governs realized flows into unemployment, while perceived job-loss risk governs the worker's search decision. When workers think job loss is more likely, an outside offer becomes more valuable as insurance, so they search more.

This mechanism matters for behavior and for the cyclicity of search. In the calibrated model, employed search rises in recessions even though labor-market tightness is lower and offers are harder to get. If the precautionary motive is removed, the tightness effect dominates and search becomes procyclical. Beliefs about job-loss risk are therefore central for why the model can generate countercyclical employed search.

The aggregate unemployment effect is smaller. In the baseline outside-option model, where most offers can insure job loss, precautionary search offsets a meaningful share of the recessionary rise in unemployment. Once the model also matches ordinary employer-to-employer mobility and the share of those moves related to job loss, the effect falls to about 0.05 percentage points in the SCE core calibration, and to about 0.12 percentage points when a larger share of offers can be used at separation. The reason is mechanical but important: search lowers unemployment only when the extra offer is available at the moment of separation. If the offer would have generated an ordinary job-to-job move anyway, it changes mobility but not unemployment.

The main contribution is therefore not that precautionary search is a large stabilizer of aggregate unemployment. It is that aggregate beliefs have a clear effect on employed workers' search decisions, and that this effect operates through perceived job-loss risk. Whether those beliefs matter much for aggregate unemployment depends on the composition of

offers that search generates. This distinction is hard to see in the survey evidence alone; the model makes it explicit by separating ordinary ladder search from search that insures against job loss.

Several caveats remain. The empirical evidence compares workers in the same time and place who report different beliefs, but it cannot fully rule out unobserved worker or job heterogeneity. The model also abstracts from congestion between employed and unemployed job seekers, which would likely reduce the aggregate unemployment effect. Better causal variation in beliefs, especially linked to administrative job transitions, would help sharpen the magnitudes. The robust conclusion is the behavioral one: when employed workers become more worried about job loss, they search more before the separation occurs.

## **A Data Appendix: Survey of Consumer Expectations (SCE)**

This appendix provides additional detail on the Survey of Consumer Expectations (SCE) and the construction of the empirical analysis sample. The main text documents that perceived job-loss risk predicts on-the-job search intensity and motivates a precautionary interpretation. Because this interpretation hinges on how expectations and search are measured, this appendix clarifies survey design, variable construction, and sample selection.

### **A.1 Survey design and panel structure**

The SCE is a nationally representative monthly internet panel administered by the Federal Reserve Bank of New York since 2013. Approximately 1,300 household heads participate each month. Respondents remain in the panel for up to 12 consecutive months, allowing expectations and realized outcomes to be linked within person. All reported statistics use SCE survey weights. Regressions cluster standard errors at the individual level to account for within-person serial correlation.

The panel structure is central for two reasons emphasized in the main text. First, it allows validation of beliefs against subsequent realizations (Section 2.4). Second, it permits specifications using lagged beliefs and within-person variation, reported below, to address concerns about contemporaneous shocks jointly driving expectations and search.

## A.2 Core survey and labor-market expectations

The monthly *core survey* elicits probabilistic expectations on a 0–100 scale. The analysis uses four measures:

1. *Macroeconomic pessimism*: the subjective probability that the U.S. unemployment rate will be higher in 12 months.
2. *Job-loss probability*: the subjective probability of involuntarily losing the current job within 12 months.
3. *Voluntary separation probability*: the subjective probability of leaving the job voluntarily within 12 months.
4. *Job-finding probability conditional on job loss*: the subjective probability of finding a job within three months if the current job is lost.

All expectations are rescaled to  $[0, 1]$ . The distinction between involuntary job-loss beliefs and voluntary separation beliefs is economically important. The model’s precautionary motive operates through perceived *inflow risk* into unemployment. The empirical analysis therefore distinguishes involuntary risk from voluntary mobility and from beliefs about re-employment prospects.

## A.3 Labor Market Module and search measurement

The *Labor Market Module* (LMM) is administered every four months to respondents while in the panel. The primary outcome variable is weekly job-search intensity: total hours spent searching or applying for jobs in the past seven days. For each LMM interview month, I match expectations from the core survey in the same calendar month (or the most recent prior core-survey response if the same-month observation is missing).

Search hours are winsorized at the 5th and 95th percentiles to limit the influence of extreme reports. The right-skewed distribution motivates robustness checks using alternative functional forms (Appendix Table A5). Importantly, the model interprets search as an intensity choice, not literal hours; accordingly, the calibration targets the *semi-elasticity* of search with respect to perceived job-loss risk rather than the level of hours.

The LMM also records industry of current employment (LMind). This variable is used to construct industry fixed effects and industry×time fixed effects in robustness specifications reported below. These specifications address the concern that unobserved industry-level shocks may jointly affect beliefs and search.

## A.4 Appendix Table A1: Summary Statistics

Appendix Table A1 reports weighted summary statistics for the employed LMM sample.

**Table A1.** Summary Statistics: Employed Workers

|                                           | N    | Mean   | SD     |
|-------------------------------------------|------|--------|--------|
| Search hours (winsorized)                 | 4226 | 3.774  | 4.800  |
| $E_t(\Delta \Pr(U_t + 12 \geq 0))$        | 4217 | 0.394  | 0.239  |
| $E_t(\Pr(\text{Lose Job}_t + 12))$        | 4220 | 0.213  | 0.234  |
| $E_t(\Pr(\text{Leave Job}_t + 12))$       | 4220 | 0.402  | 0.304  |
| $E_t(\Pr(\text{Find} \text{Lose}_t + 3))$ | 4219 | 0.571  | 0.300  |
| $E_t(\pi_t + 12)$ (point)                 | 4219 | 0.072  | 0.213  |
| Age                                       | 4226 | 41.508 | 11.911 |
| Female                                    | 4224 | 0.453  | 0.498  |
| Married                                   | 4225 | 0.599  | 0.490  |
| White                                     | 4226 | 0.772  | 0.419  |
| Black                                     | 4226 | 0.145  | 0.352  |
| Hispanic                                  | 4225 | 0.120  | 0.325  |
| HH income: Under 50k                      | 4211 | 0.376  | 0.484  |
| HH income: 50k–100k                       | 4211 | 0.324  | 0.468  |
| HH income: Over 100k                      | 4211 | 0.300  | 0.459  |
| Education: High school                    | 4205 | 0.240  | 0.427  |
| Education: Some college                   | 4205 | 0.293  | 0.455  |
| Education: College+                       | 4205 | 0.467  | 0.499  |

Sample: Employed workers with non-missing search hours (SCE Labor Market Module).

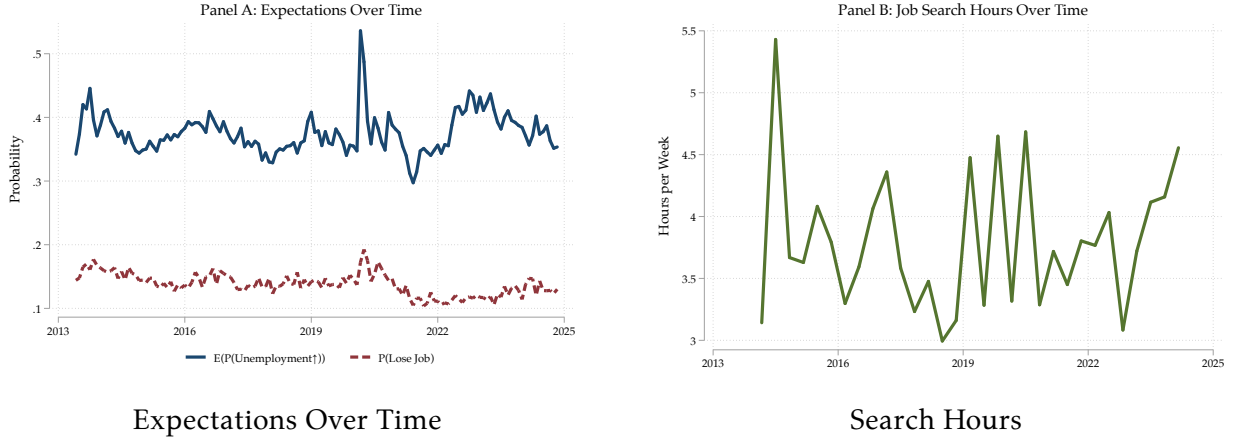
Search hours winsorized at 5th and 95th percentiles.

Means/SD computed with SCE weights (analytic weights); N is unweighted non-missing count.

Binary indicators report sample shares (means).

Two features are worth emphasizing. First, subjective expectations display substantial dispersion. The cross-sectional standard deviation of job-loss beliefs is comparable to its mean. This dispersion underlies the main empirical design: within a given state and month, workers hold different beliefs about separation risk.

Second, search intensity is highly skewed. The mean is approximately 3.8 hours per week, but the standard deviation exceeds the mean. This motivates both winsorization and the use of alternative specifications in Appendix Table A5. The empirical results in the main text do not depend on functional form.



Notes: The figure plots weighted monthly means of macro unemployment expectations (core survey) and weekly search hours (LMM) for employed respondents. Expectations spike during major aggregate events (notably COVID-19). Search hours are more volatile because the LMM is only administered every four months to a rotating subsample. The aggregate time series is therefore noisy, motivating the use of within-state-month cross-sectional variation in the main analysis.

**Figure A1.** Time Series of Expectations and Search Intensity

## A.5 Appendix Figure A1: Aggregate Time Series

# B Empirical Appendix

This section collects robustness and identification checks referenced in Section 2. Each exhibit addresses a specific concern about the interpretation of the belief–search relationship. Throughout, the central question is whether the positive association between perceived job-loss risk and on-the-job search intensity reflects a precautionary response to perceived separation risk, or instead captures omitted local shocks, sectoral conditions, functional-form artifacts, or stable individual heterogeneity.

## B.1 Alternative Fixed Effects

The specification replaces separate state and time fixed effects with their full interaction:

$$\text{SearchHours}_{ist} = \beta' \mathbf{E}_{it} + \mathbf{X}_{it}' \gamma + \mu_{st} + \varepsilon_{ist}, \quad (10)$$

where  $\mathbf{E}_{it}$  collects the expectation variables (added progressively across columns),  $\mathbf{X}_{it}$  are demographic controls, and  $\mu_{st}$  denotes state×time fixed effects. Standard errors are clustered at the individual level; observations are weighted by survey weights. The sample consists of employed workers.

**Table A2.** Expectations and On-the-Job Search Intensity

|                                                                                | (1)                 | (2)                 | (3)                  | (4)                  | (5)                 |
|--------------------------------------------------------------------------------|---------------------|---------------------|----------------------|----------------------|---------------------|
|                                                                                | Baseline            | + Separation        | + P(find)            | + P(offer)           | Interaction         |
| $E_t[\Delta \Pr(U_{t+12} \geq 0)]$                                             | 1.538***<br>(0.420) | 0.920**<br>(0.415)  | 0.906**<br>(0.412)   | 0.814*<br>(0.435)    | 0.870**<br>(0.408)  |
| $E_t \Pr(\text{Lose Job}_{t+12})$                                              |                     | 2.064***<br>(0.499) | 1.841***<br>(0.502)  | 1.533***<br>(0.517)  | 2.719***<br>(0.991) |
| $E_t \Pr(\text{Leave Job}_{t+12})$                                             |                     | 1.372***<br>(0.398) | 1.691***<br>(0.407)  | 1.040**<br>(0.439)   | 1.692***<br>(0.407) |
| $E_t \Pr(\text{Find Job}_{t+3})$                                               |                     |                     | -1.259***<br>(0.352) | -1.762***<br>(0.360) | -0.973**<br>(0.441) |
| $E_t \Pr(\text{Offer}_{t+3})$                                                  |                     |                     |                      | 0.020***<br>(0.004)  |                     |
| $E_t(\Pr(\text{Lose Job}_{t+12})) \times$<br>$E_t(\Pr(\text{Find Job}_{t+3}))$ |                     |                     |                      |                      | -1.523<br>(1.452)   |
| $E_t(\pi_{t+12})$                                                              | -0.019<br>(0.023)   | -0.019<br>(0.023)   | -0.023<br>(0.023)    | -0.019<br>(0.024)    | -0.024<br>(0.023)   |
| Separation expectations                                                        | No                  | Yes                 | Yes<br>P(find)       | Yes                  | Yes                 |
| Job-finding expectations                                                       | No                  | No                  | given lose           | Both                 | Interac-<br>tion    |
| State $\times$ Time FE                                                         | Yes                 | Yes                 | Yes                  | Yes                  | Yes                 |
| Observations                                                                   | 4132                | 4131                | 4130                 | 3922                 | 4130                |
| R-squared                                                                      | 0.391               | 0.405               | 0.409                | 0.419                | 0.409               |

Notes: Dependent variable: weekly search hours (winsorized). Expectations scaled to  $[0, 1]$ . Specification absorbs state $\times$ time fixed effects; controls include age polynomial, education, race, gender, marital status, household income; weighted by survey weights; standard errors clustered by individual; sample: employed workers. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**State $\times$ time fixed effects.** This specification absorbs all state-level labor-market conditions in a given month, including unemployment rates, vacancy conditions, and other local shocks. Identification therefore comes entirely from cross-sectional differences in beliefs among workers residing in the *same state in the same month*.

The stability of the job-loss coefficient relative to the baseline indicates that the belief–search relationship is not driven by common local shocks. Even within a fixed local environment, workers who perceive higher separation risk search more intensively.

This specification replaces time fixed effects with industry $\times$ time interactions and retains



state fixed effects:

$$\text{SearchHours}_{ijst} = \beta' \mathbf{E}_{it} + \mathbf{X}'_{it} \gamma + \mu_s + \delta_{jt} + \varepsilon_{ijst},$$

where  $\delta_{jt}$  denotes industry×time fixed effects (using the LMM industry classification) and  $\mu_s$  are state fixed effects.

**Industry×time fixed effects.** Industry×time fixed effects absorb sector-specific shocks at a monthly frequency, addressing the concern that industry downturns jointly raise perceived risk and search.

The job-loss coefficient remains economically and statistically meaningful under this demanding specification. This indicates that the belief–search relationship is not merely reflecting industry-wide conditions, but also heterogeneity in perceived risk among workers in the same industry at the same point in time.

The sample is restricted to employed workers with non-missing industry. Column (1) uses state×time fixed effects ( $\mu_{st}$ ); Column (2) adds industry fixed effects ( $\delta_i$ ); Column (3) replaces these with industry×time fixed effects ( $\delta_{jt}$ ) alongside state fixed effects ( $\mu_s$ ).

**Industry and state controls combined.** Adding industry fixed effects alongside time and state controls further isolates within-industry, within-state belief variation. The magnitude and sign of the key coefficients are preserved. Taken together, these specifications show that the precautionary relationship survives controls for both geographic and sectoral conditions.

Importantly, these exercises do not establish causality. They show that the empirical relationship is not explained by observable local or sectoral shocks, strengthening the case that belief heterogeneity plays an independent role.

## B.2 Alternative Outcomes and Timing

Columns (1)–(3) estimate the baseline specification with alternative dependent variables:

$$f(\text{SearchHours}_{ist}) = \beta' \mathbf{E}_{it} + \mathbf{X}'_{it} \gamma + \mu_s + \lambda_t + \varepsilon_{ist},$$

where  $f(\cdot)$  is the identity (levels),  $\log(1 + \cdot)$ , or the arc-percent transformation  $(y - \bar{y}) / (0.5(y + \bar{y}))$ , with  $\bar{y}$  denoting mean search hours in the estimation sample. Column (4) estimates a Poisson quasi-maximum likelihood model with the same regressors and fixed effects.

**Table A3.** Expectations and On-the-Job Search Intensity

|                                                                              | (1)<br>Baseline     | (2)<br>+ Separation | (3)<br>+ P(find)      | (4)<br>+ P(offer)    | (5)<br>Interaction   |
|------------------------------------------------------------------------------|---------------------|---------------------|-----------------------|----------------------|----------------------|
| $E_t$<br>$\Delta \Pr(U_{t+12} \geq 0)$                                       | 1.671***<br>(0.452) | 0.981**<br>(0.435)  | 1.010**<br>(0.435)    | 1.110**<br>(0.454)   | 1.006**<br>(0.432)   |
| $E_t$<br>$\Pr(\text{Lose Job}_{t+12})$                                       |                     | 2.107***<br>(0.483) | 1.844***<br>(0.493)   | 1.652***<br>(0.515)  | 1.976**<br>(0.975)   |
| $E_t$<br>$\Pr(\text{Leave Job}_{t+12})$                                      |                     | 1.595***<br>(0.369) | 1.917***<br>(0.379)   | 1.411***<br>(0.415)  | 1.917***<br>(0.379)  |
| $E_t$<br>$\Pr(\text{Find Job}_{t+3})$                                        |                     |                     | -1.222***<br>(0.342)  | -1.566***<br>(0.358) | -1.176***<br>(0.441) |
| $E_t$<br>$\Pr(\text{Offer}_{t+3})$                                           |                     |                     |                       | 0.014***<br>(0.004)  |                      |
| $E_t \Pr(\text{Lose Job}_{t+12})$<br>$\times E_t \Pr(\text{Find Job}_{t+3})$ |                     |                     |                       |                      | -0.227<br>(1.432)    |
| $E_t(\pi_{t+12})$<br>(dist. mean)                                            | -0.014<br>(0.027)   | -0.019<br>(0.027)   | -0.023<br>(0.027)     | -0.020<br>(0.029)    | -0.023<br>(0.027)    |
| Separation expectations                                                      | No                  | Yes                 |                       | Yes                  | Yes                  |
| Job-finding expectations                                                     | No                  | No                  | P(find)<br>given lose | Both                 | Interac-<br>tion     |
| Industry $\times$ Time FE                                                    | Yes                 | Yes                 | Yes                   | Yes                  | Yes                  |
| State FE                                                                     | Yes                 | Yes                 | Yes                   | Yes                  | Yes                  |
| Observations                                                                 | 3086                | 3085                | 3085                  | 2925                 | 3085                 |
| R-squared                                                                    | 0.332               | 0.350               | 0.354                 | 0.361                | 0.354                |

Notes: Dependent variable: weekly search hours (winsorized). Expectations scaled to  $[0, 1]$ . Specification absorbs industry $\times$ time fixed effects and includes state FE; controls include age polynomial, education, race, gender, marital status, household income; weighted by survey weights; standard errors clustered by individual; sample: employed workers. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Table A4.** Main Result Robustness: Industry Controls (LMind)

|                                                                              | (1)<br>State×Time FE | (2)<br>+ Industry FE | (3)<br>Industry×Time FE |
|------------------------------------------------------------------------------|----------------------|----------------------|-------------------------|
| $E_t[\Delta \Pr(U_{t+12} \geq 0)]$                                           | 0.853*<br>(0.499)    | 0.772<br>(0.508)     | 1.006**<br>(0.432)      |
| $E_t \Pr(\text{Lose Job}_{t+12})$                                            | 1.310<br>(1.152)     | 1.379<br>(1.161)     | 1.976**<br>(0.975)      |
| $E_t \Pr(\text{Leave Job}_{t+12})$                                           | 2.104***<br>(0.464)  | 2.068***<br>(0.464)  | 1.917***<br>(0.379)     |
| $E_t \Pr(\text{Find Job}_{t+3})$                                             | -1.433**<br>(0.588)  | -1.356**<br>(0.575)  | -1.176***<br>(0.441)    |
| $E_t \Pr(\text{Lose Job}_{t+12}) \times$<br>$E_t \Pr(\text{Find Job}_{t+3})$ | 0.466<br>(1.775)     | 0.460<br>(1.769)     | -0.227<br>(1.432)       |
| $E_t(\pi_{t+12})$                                                            | -0.017<br>(0.029)    | -0.014<br>(0.029)    | -0.023<br>(0.027)       |
| State × Time FE                                                              | Yes                  | Yes                  | No                      |
| Industry × Time FE                                                           | No                   | No                   | Yes                     |
| Industry FE                                                                  | No                   | Yes                  | No                      |
| Observations                                                                 | 3085                 | 3085                 | 3085                    |
| R-squared                                                                    | 0.489                | 0.495                | 0.354                   |

Notes: Sample restricted to employed workers with non-missing LMind industry. Column 1 absorbs state×time FE; column 2 adds industry FE; column 3 absorbs industry×time FE and includes state FE. Expectations scaled to  $[0, 1]$ ; weighted by survey weights; standard errors clustered by individual. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Functional form robustness.** Search hours are highly skewed, with a substantial mass at zero. Appendix Table A5 shows that the positive association between perceived job-loss probability and search intensity is robust across alternative transformations and specifications, including  $\log(1 + y)$ , the arc-percent transformation, and Poisson quasi-maximum likelihood.

The dependent variable is an indicator for any positive search hours:

$$\mathbb{1}\{\text{SearchHours}_{ist} > 0\} = \beta' \mathbf{E}_{it} + \mathbf{X}'_{it} \gamma + \mu_s + \lambda_t + \varepsilon_{ist}.$$

**Extensive margin.** Appendix Table A6 shows that perceived job-loss probability significantly predicts the probability of engaging in any search at all. Macro pessimism loses statistical significance once personal beliefs are included, consistent with the separation-risk-first interpretation emphasized in the main text.

Contemporaneous expectations are replaced with their values from the most recent

**Table A5.** Robustness: Alternative Dependent Variable Specifications

|                                           | (1)<br>Levels        | (2)<br>Log           | (3)<br>Arc-%         | (4)<br>Poisson       |
|-------------------------------------------|----------------------|----------------------|----------------------|----------------------|
| main                                      |                      |                      |                      |                      |
| $E_t(\Delta \Pr(U_t + 12 \geq 0))$        | 0.953***<br>(0.355)  | 0.130**<br>(0.058)   | 0.144**<br>(0.070)   | 0.246***<br>(0.091)  |
| $E_t(\Pr(\text{Lose Job}_t + 12))$        | 1.836***<br>(0.429)  | 0.344***<br>(0.065)  | 0.413***<br>(0.077)  | 0.398***<br>(0.094)  |
| $E_t(\Pr(\text{Leave Job}_t + 12))$       | 1.884***<br>(0.333)  | 0.389***<br>(0.052)  | 0.479***<br>(0.062)  | 0.512***<br>(0.083)  |
| $E_t(\Pr(\text{Find} \text{Lose}_t + 3))$ | -1.156***<br>(0.294) | -0.210***<br>(0.050) | -0.251***<br>(0.060) | -0.351***<br>(0.078) |
| Dep. Variable                             | Levels               | Log(1+y)             | Arc-%                | Poisson              |
| Observations                              | 4130                 | 4130                 | 4130                 | 4130                 |
| R-squared                                 | 0.152                | 0.159                | 0.156                |                      |

Notes: Dependent variable variants: levels, log(1+y), arc-%  $(y - \bar{y}) / (0.5(y + \bar{y}))$ , and Poisson. Expectations scaled to [0, 1]. All specifications include state and time fixed effects plus controls; weighted by survey weights; standard errors clustered by individual (Poisson uses robust SE). \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

prior core-survey wave:

$$\text{SearchHours}_{ist} = \beta' \mathbf{E}_{i,t-1} + \mathbf{X}'_{it} \gamma + \mu_s + \lambda_t + \varepsilon_{ist},$$

where  $\mathbf{E}_{i,t-1}$  denotes expectations measured before the LMM interview.

**Lagged expectations and timing.** Appendix Table A7 replaces contemporaneous beliefs with beliefs reported in a prior core-survey wave that precedes the LMM interview. The persistence of the job-loss coefficient under lagged beliefs suggests that the belief–search relationship is not purely contemporaneous. While this does not fully resolve reverse-causality concerns, it indicates that beliefs measured before the search observation predict subsequent search intensity.

Two within-person specifications are estimated. First, a first-difference specification:

$$\Delta \text{SearchHours}_{it} = \beta' \Delta \mathbf{E}_{i,t-1} + \lambda_t + \Delta \varepsilon_{it}.$$

Second, an individual fixed-effects specification using lagged expectations:

$$\text{SearchHours}_{it} = \beta' \mathbf{E}_{i,t-1} + \alpha_i + \lambda_t + \varepsilon_{it},$$

**Table A6.** Expectations and On-the-Job Search: Extensive Margin

|                                                                              | (1)<br>Baseline   | (2)<br>+ Separation | (3)<br>+ P(find)    | (4)<br>+ P(offer)   | (5)<br>Interaction  |
|------------------------------------------------------------------------------|-------------------|---------------------|---------------------|---------------------|---------------------|
| $E_t$<br>$\Delta \Pr(U_{t+12} \geq 0)$                                       | 0.039<br>(0.026)  | 0.014<br>(0.026)    | 0.015<br>(0.026)    | 0.003<br>(0.026)    | 0.013<br>(0.026)    |
| $E_t$<br>$\Pr(\text{Lose Job}_{t+12})$                                       |                   | 0.072***<br>(0.024) | 0.066***<br>(0.024) | 0.057**<br>(0.025)  | 0.101**<br>(0.051)  |
| $E_t$<br>$\Pr(\text{Leave Job}_{t+12})$                                      |                   | 0.064***<br>(0.021) | 0.073***<br>(0.022) | 0.040<br>(0.024)    | 0.073***<br>(0.022) |
| $E_t$<br>$\Pr(\text{Find Job}_{t+3})$                                        |                   |                     | -0.033<br>(0.023)   | -0.054**<br>(0.023) | -0.022<br>(0.029)   |
| $E_t$<br>$\Pr(\text{Offer}_{t+3})$                                           |                   |                     |                     | 0.001***<br>(0.000) |                     |
| $E_t \Pr(\text{Lose Job}_{t+12})$<br>$\times E_t \Pr(\text{Find Job}_{t+3})$ |                   |                     |                     |                     | -0.062<br>(0.081)   |
| $E_t(\pi_{t+12})$<br>(dist. mean)                                            | -0.002<br>(0.001) | -0.002<br>(0.001)   | -0.002<br>(0.001)   | -0.002<br>(0.001)   | -0.002<br>(0.001)   |
| Separation expectations                                                      | No                | Yes                 | Yes<br>P(find)      | Yes                 | Yes                 |
| Job-finding expectations                                                     | No                | No                  | given lose          | Both                | Interac-<br>tion    |
| Time FE                                                                      | Yes               | Yes                 | Yes                 | Yes                 | Yes                 |
| State FE                                                                     | Yes               | Yes                 | Yes                 | Yes                 | Yes                 |
| Observations                                                                 | 4132              | 4131                | 4130                | 3922                | 4130                |
| R-squared                                                                    | 0.045             | 0.051               | 0.052               | 0.061               | 0.052               |

Notes: Dependent variable: indicator for any search ( $> 0$  hours). Expectations scaled to  $[0, 1]$ . State and time fixed effects with demographic controls; weighted by survey weights; standard errors clustered by individual; sample: employed workers. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Table A7.** Lagged Expectations and On-the-Job Search Intensity

|                                                                                                            | (1)                 | (2)                 | (3)                  | (4)                  |
|------------------------------------------------------------------------------------------------------------|---------------------|---------------------|----------------------|----------------------|
|                                                                                                            | Baseline            | + Separation        | + P(find)            | Interaction          |
| Lagged $E_t(\Delta \Pr(U_{-t} + 12 \geq 0))$                                                               | 1.322***<br>(0.405) | 0.453<br>(0.428)    | 0.419<br>(0.429)     | 0.384<br>(0.429)     |
| Lagged $E_t(\Pr(\text{Lose Job}_{-t} + 12))$                                                               |                     | 2.323***<br>(0.484) | 2.097***<br>(0.490)  | 2.918***<br>(0.892)  |
| Lagged $E_t(\Pr(\text{Leave Job}_{-t} + 12))$                                                              |                     | 1.483***<br>(0.334) | 1.751***<br>(0.336)  | 1.773***<br>(0.337)  |
| Lagged $E_t(\Pr(\text{Find} \text{Lose}_{-t} + 3))$                                                        |                     |                     | -1.201***<br>(0.292) | -0.948***<br>(0.336) |
| Lagged $E_t(\Pr(\text{Offer}_{-t} + 3))$                                                                   |                     |                     |                      |                      |
| Lagged $E_t(\Pr(\text{Lose Job}_{-t} + 12)) \times$<br>Lagged $E_t(\Pr(\text{Find} \text{Lose}_{-t} + 3))$ |                     |                     |                      | -1.444<br>(1.336)    |
| Lagged $E_t(\pi_{-t} + 12)$ (dist. mean)                                                                   | 0.014<br>(0.021)    | 0.020<br>(0.022)    | 0.018<br>(0.022)     | 0.018<br>(0.022)     |
| Separation expectations                                                                                    | No                  | Yes                 | Yes<br>P(find)       | Yes                  |
| Job-finding expectations                                                                                   | No                  | No                  | given lose           | Interac-<br>tion     |
| Time FE                                                                                                    | Yes                 | Yes                 | Yes                  | Yes                  |
| State FE                                                                                                   | Yes                 | Yes                 | Yes                  | Yes                  |
| Observations                                                                                               | 4133                | 3975                | 3974                 | 3974                 |
| R-squared                                                                                                  | 0.122               | 0.139               | 0.144                | 0.145                |

Notes: Lagged expectations from the prior core survey wave ( $\approx 4$  months earlier). Dependent variable: weekly search hours (winsorized). State and time fixed effects with demographic controls; weighted by survey weights; standard errors clustered by individual; sample: employed workers. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

where  $\alpha_i$  denotes individual fixed effects. Standard errors are clustered at the individual level.

**Table A8.** Within-Person Identification: First Differences and Fixed Effects

|                                           | (1)              | (2)                 | (3)              | (4)                 |
|-------------------------------------------|------------------|---------------------|------------------|---------------------|
|                                           | FD Base          | FD + Sep.           | FE Base          | FE + Sep.           |
| $\Delta E_t[\Delta \Pr(U_{t+12} \geq 0)]$ | 1.335<br>(0.957) | 0.905<br>(0.918)    |                  |                     |
| Lagged $E_t[\Delta \Pr(U_{t+12} \geq 0)]$ |                  |                     | 0.404<br>(0.543) | -0.201<br>(0.558)   |
| $\Delta E_t \Pr(\text{Lose Job}_{t+12})$  |                  | 1.334<br>(1.034)    |                  |                     |
| Lagged $E_t \Pr(\text{Lose Job}_{t+12})$  |                  |                     |                  | 1.103<br>(0.834)    |
| $\Delta E_t \Pr(\text{Leave Job}_{t+12})$ |                  | 2.450***<br>(0.831) |                  |                     |
| Lagged $E_t \Pr(\text{Leave Job}_{t+12})$ |                  |                     |                  | 1.875***<br>(0.514) |
| Specification                             | First Diff       | First Diff          | Individual FE    | Individual FE       |
| Observations                              | 1183             | 1111                | 4576             | 4402                |
| R-squared                                 | 0.069            | 0.079               | 0.029            | 0.045               |

Notes: Columns 1–2 use first differences; columns 3–4 include individual fixed effects. Expectations scaled to  $[0, 1]$ . Time controls as in main specs; standard errors clustered by individual; short panel (average 1.4 LMM observations per worker) limits within-person power. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Within-person variation.** Individual fixed-effects and first-difference specifications exploit within-person variation in beliefs over time. Statistical power is limited because most respondents appear only once or twice in the LMM. Consistent with this limitation, the within-person estimates are imprecise. They are nonetheless broadly consistent in sign with the cross-sectional evidence, particularly for voluntary separation beliefs.

### B.3 Employed vs. Unemployed and Heterogeneity

The baseline specification is estimated separately for employed and unemployed workers. For the unemployed, the expectation vector includes macro pessimism and perceived

job-finding probability (job-loss beliefs are not applicable):

$$\text{SearchHours}_{ist} = \beta' \mathbf{E}_{it} + \mathbf{X}'_{it} \gamma + \mu_s + \lambda_t + \varepsilon_{ist}.$$

**Table A9.** Expectations and Search: Employed vs. Unemployed Workers

|                                            | (1)<br>Employed      | (2)<br>Unemployed   |
|--------------------------------------------|----------------------|---------------------|
| $E_t(\Delta \Pr(U_t + 12 \geq 0))$         | 0.953***<br>(0.355)  | 0.181<br>(1.318)    |
| $E_t(\Pr(\text{Lose Job}_t + 12))$         | 1.836***<br>(0.429)  |                     |
| $E_t(\Pr(\text{Leave Job}_t + 12))$        | 1.884***<br>(0.333)  |                     |
| $E_t(\Pr(\text{Find} \text{Lose}_t + 3))$  | -1.156***<br>(0.294) |                     |
| $E_t(\Pr(\text{Find} \text{Unemp}_t + 3))$ |                      | 2.130**<br>(1.018)  |
| $E_t(\pi_t + 12)$ (dist. mean)             | -0.028<br>(0.022)    | 0.134***<br>(0.039) |
| Sample                                     | Employed             | Unemployed          |
| Observations                               | 4130                 | 712                 |
| R-squared                                  | 0.152                | 0.295               |

Notes: Employed sample: search hours on separation and job-finding expectations; unemployed sample: search hours on macro expectations and job-finding probability. Expectations scaled to [0, 1]. State and time fixed effects with demographic controls; weighted by survey weights; standard errors clustered by individual. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Employment status comparison.** Appendix Table A9 compares belief–search relationships across employment status. The belief–search relationship is strongest among employed workers, consistent with the interpretation that on-the-job search serves an insurance role against separation risk.

The baseline specification is estimated on subsamples defined by marital status, median income split, and age (< 40 vs. ≥ 40).



**Table A10.** Heterogeneity by Marital Status, Income, and Age

|                                    | (1)                  | (2)                  | (3)                  | (4)                 | (5)                 | (6)                  |
|------------------------------------|----------------------|----------------------|----------------------|---------------------|---------------------|----------------------|
|                                    | Unmarried            | Married              | Low Inc.             | High Inc.           | Young               | Old                  |
| $E_t[\Delta \Pr(U_{t+12} \geq 0)]$ | 1.125*<br>(0.658)    | 0.887**<br>(0.408)   | 1.254***<br>(0.467)  | -0.030<br>(0.464)   | 1.272**<br>(0.516)  | 0.647<br>(0.483)     |
| $E_t \Pr(\text{Lose Job}_{t+12})$  | 2.525***<br>(0.672)  | 0.843<br>(0.552)     | 1.416**<br>(0.572)   | 2.243***<br>(0.628) | 1.587**<br>(0.696)  | 1.906***<br>(0.578)  |
| $E_t \Pr(\text{Leave Job}_{t+12})$ | 1.698***<br>(0.557)  | 2.311***<br>(0.419)  | 2.100***<br>(0.462)  | 1.723***<br>(0.442) | 2.090***<br>(0.502) | 1.985***<br>(0.473)  |
| $E_t \Pr(\text{Find Job}_{t+3})$   | -1.173***<br>(0.455) | -1.298***<br>(0.402) | -1.366***<br>(0.400) | -0.510<br>(0.401)   | -0.862*<br>(0.466)  | -1.449***<br>(0.374) |
| Sample                             | Unmarried            | Married              | Low Income           | High Income         | Age < 40            | Age ≥ 40             |
| Observations                       | 1675                 | 2512                 | 2844                 | 1343                | 1989                | 2198                 |
| R-squared                          | 0.221                | 0.189                | 0.164                | 0.139               | 0.187               | 0.207                |

Notes: Subsamples by marital status, income (median split), and age (< 40 vs ≥ 40). Expectations scaled to [0, 1]. State and time fixed effects with demographic controls; weighted by survey weights; standard errors clustered by individual. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Worker heterogeneity.** Appendix Table A10 explores heterogeneity by marital status, income, and age. The belief–search relationship is present across subgroups and is not confined to a narrow segment of the labor market.

## B.4 Additional Validation of the Separation-Risk Channel

To further assess whether subjective job-loss beliefs contain information about realized labor-market risk, this subsection reports two additional validation exercises that are not used for identification in the main search regressions.

**Table A11.** Validation: Beliefs Predict Separations at Multiple Horizons (Q10-based)

|                      | (1)                   | (2)                   | (3)                   | (4)                   | (5)                                     | (6)                                | (7)                                |
|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------------------------|------------------------------------|------------------------------------|
|                      | 1m                    | 3m                    | 6m                    | 9m                    | 12m exact                               | Any<=12 (fup>=1)                   | Any<=12 (fup>=6)                   |
| $P(\text{Lose Job})$ | 0.0770***<br>(0.0065) | 0.1290***<br>(0.0107) | 0.1487***<br>(0.0156) | 0.1266***<br>(0.0198) | 0.5101<br>(0.3434)                      | 0.1996***<br>(0.0159)              | 0.2504***<br>(0.0245)              |
| Outcome              | Not employed/attached | Not employed/attached | Not employed/attached | Not employed/attached | Not employed/attached (exact; selected) | Any not-employed (censored), fup≥1 | Any not-employed (censored), fup≥6 |
| Horizon              | 1 month               | 3 months              | 6 months              | 9 months              | 12 months                               | ≤12 months                         | ≤12 months                         |
| Observations         | 71116                 | 53832                 | 34656                 | 16586                 | 290                                     | 71116                              | 27032                              |
| R-squared            | 0.171                 | 0.205                 | 0.287                 | 0.415                 | 0.690                                   | 0.206                              | 0.349                              |

Exact-horizon outcomes require observation at t+h.

Any<=12 is right-censored and uses observed forward months; sample restrictions based on consecutive follow-up fup\_consec12.

**Prediction at multiple horizons.** Appendix Table A11 re-estimates the validation specification at horizons from 1 to 9 months and with two right-censored outcomes for “any non-employment within 12 months.” The coefficient on  $P(\text{Lose Job})$  is positive and precisely estimated at short and medium horizons, and remains positive in the censored 12-month outcomes. This pattern indicates that the belief measure tracks realized separation-related risk over horizons that are relevant for employed search decisions.

**Table A12.** Robustness: Beliefs Predict Both Non-Employment and Unemployment (Q10-based)

|                      | (1)                         | (2)                   |
|----------------------|-----------------------------|-----------------------|
|                      | Not employed/attached (t+3) | Unemployed (t+3)      |
| $P(\text{Lose Job})$ | 0.1290***<br>(0.0107)       | 0.0726***<br>(0.0079) |
| Observations         | 55832                       | 55832                 |
| R-squared            | 0.205                       | 0.186                 |

**Outcome-definition robustness.** Appendix Table A12 compares two  $t+3$  outcomes: broad non-employment/attachment and strict unemployment. The job-loss belief coefficient is positive and statistically significant in both columns. The precautionary interpretation therefore does not depend on one narrow coding of post-separation labor-market status.

## B.5 ECB Consumer Expectations Survey Validation

Appendix Table A13 reports external-validation exercises in the ECB Consumer Expectations Survey, using the April 2026 release through 2026Q1. Panels A and B use employed respondents in the regular quarterly labor-market module matched to monthly expectations. Panel A asks whether aggregate labor-market pessimism predicts perceived job-loss risk. Panel B asks whether perceived job-loss risk predicts search outcomes. Panel C reports a supplementary heterogeneity check. The regular quarterly module does not observe an employed-worker job-finding probability, so I use the topical labor-market module's expected time to find an equivalent job and merge it forward to later quarterly labor-market observations for the same respondent. Restricting this merge to the nearest subsequent quarterly round gives the same qualitative conclusion.

## B.6 2016 Election Event Study

This subsection provides an illustration of one source of within-location belief dispersion: partisan priors interacting with salient political events. Following Mian et al. (2018) and Coibion et al. (2020), the 2016 U.S. presidential election generated a sharp divergence in macroeconomic expectations across partisan-leaning groups. I use this episode to show that workers in similar local labor markets can hold systematically different macro views, and that these views can shift discontinuously around a major news event. I emphasize that this exercise is *illustrative*: it is not used as a stand-alone identification strategy for the causal effect of beliefs on search.

**Table A13.** External Validation in the ECB Consumer Expectations Survey

| <b>Panel A. Macro pessimism predicts perceived job-loss risk</b>                       |                                        |                               |         |
|----------------------------------------------------------------------------------------|----------------------------------------|-------------------------------|---------|
| Outcome                                                                                | Coef. on expected unemployment         | SE                            | N       |
| <i>P</i> (lose job in next 3 months)                                                   | 0.1450***                              | (0.0104)                      | 212,582 |
| <i>P</i> (lose job in next 3 months), with expectation controls                        | 0.0887***                              | (0.0185)                      | 212,581 |
| <b>Panel B. Perceived job-loss risk predicts search</b>                                |                                        |                               |         |
| Outcome                                                                                | Coef. on <i>P</i> (lose job)           | SE                            | N       |
| Actively looking for a job                                                             | 0.4736***                              | (0.0089)                      | 208,018 |
| <i>P</i> (look for job in next 3 months)                                               | 0.6161***                              | (0.0077)                      | 176,293 |
| Job-search intensity                                                                   | 0.3093***                              | (0.0313)                      | 25,711  |
| <b>Panel C. Supplementary interaction with expected time to find an equivalent job</b> |                                        |                               |         |
| Outcome                                                                                | Job-loss slope, stronger-finding group | Interaction with weak finding | N       |
| Actively looking for a job                                                             | 0.4367***<br>(0.0308)                  | -0.0306<br>(0.0504)           | 31,151  |
| <i>P</i> (look for job in next 3 months)                                               | 0.6229***<br>(0.0272)                  | -0.0289<br>(0.0438)           | 26,730  |
| Job-search intensity                                                                   | 0.4288***<br>(0.1183)                  | -0.5662***<br>(0.1987)        | 3,501   |

Notes: ECB CES probability variables are scaled to [0, 1]. The sample uses the ECB release dated 28 April 2026 and includes quarterly labor-market observations through 2026Q1 matched to monthly expectations. In Panel A, macro pessimism is the expected unemployment rate over the next 12 months. The second row adds controls for expected inflation, expected economic growth, current perceived unemployment, and expected household financial conditions. In Panel B, job-loss risk is the perceived probability of losing the current job in the next 3 months; these regressions also control for expected unemployment. Panels A and B use employed respondents in the regular quarterly labor-market module. Panel C uses the topical labor-market module's expected time to find an equivalent job, measured in round 41, merged forward to later regular quarterly rounds for the same respondent; it includes the same expectation controls as Panel B. "Weak finding" equals one if expected time to find an equivalent job is above the weighted median, i.e. 7 months or longer. All specifications use ECB CES weights, survey-round fixed effects, country fixed effects, demographic controls, and respondent-clustered standard errors. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Design and sample.** I classify states by whether the Republican or Democratic candidate won the state in November 2016. To focus on states with clearer partisan alignment, I drop swing states (defined based on the election margin threshold used in the replication code) and restrict attention to the 2016–2017 window. Because treatment is defined at the state level and the effective number of clusters is limited, results should be interpreted cautiously.

**Event-study specification.** I estimate an event-study that compares Republican-won and Democratic-won states around the election:

$$y_{ist} = \sum_{k \neq -1} \delta_k [\mathbb{1}\{t = k\} \times \text{RepWon}_s] + \mu_s + \lambda_t + \mathbf{X}'_{it}\gamma + \varepsilon_{ist}, \quad (11)$$

where  $t$  indexes event time (months relative to November 2016),  $\mu_s$  and  $\lambda_t$  are state and month fixed effects, and  $\mathbf{X}_{it}$  includes the baseline demographic controls. Standard errors are clustered at the state level.

**Findings.** Appendix Figure A3 shows two patterns. First, macro unemployment expectations diverge sharply after the election: respondents in Republican-won states report a persistent decline in macro pessimism relative to respondents in Democratic-won states, with little evidence of differential pre-trends. This illustrates that belief dispersion about aggregate conditions can be substantial even among workers facing similar local environments.

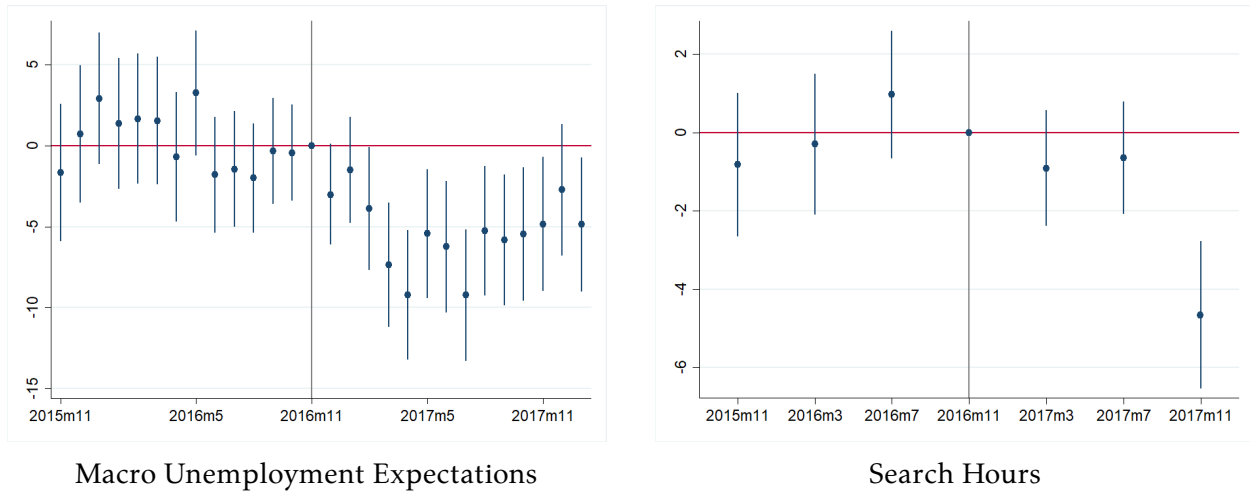
Second, the corresponding series for search hours is consistent with lower post-election search in Democrat-won states relative to Republican-won states, but is imprecisely estimated. Precision is limited because search hours are observed only in the LMM (every four months) and because the effective sample size is smaller once restricting to non-swing states. I therefore treat the search evidence as suggestive.

The difference-in-differences specification is

$$y_{ist} = \delta (\text{RepWon}_s \times \text{Post}_t) + \mathbf{X}'_{it}\gamma + \mu_s + \lambda_t + \varepsilon_{ist},$$

where  $\text{Post}_t = \mathbb{1}\{t \geq \text{Nov 2016}\}$  and the sample is restricted to non-swing states over 2016–2017. Standard errors are clustered at the state level.

The difference-in-differences specification confirms a sharp post-election divergence in macro unemployment expectations across Republican- and Democratic-won states. For search, the positive Rep-won $\times$ Post estimate implies lower post-election search in



Notes: The figure plots event-study coefficients comparing Republican- and Democrat-won non-swing states around the November 2016 election. The left panel shows divergence in macro unemployment expectations; the right panel shows corresponding movements in search hours. Outcomes are weighted; specifications include state and month fixed effects. The quarterly LMM limits precision, so this evidence is illustrative rather than definitive.

**Figure A3.** Event Study: 2016 Presidential Election

**Table A14.** 2016 Election (Appendix): DID (With Controls)

|                       | (1)                  | (2)              | (3)                | (4)              | (5)                 |
|-----------------------|----------------------|------------------|--------------------|------------------|---------------------|
|                       | E(U exp)             | P(Lose)          | P(Leave)           | Search (arc)     | Search + beliefs    |
| Rep-won $\times$ Post | -0.060***<br>(0.014) | 0.014<br>(0.010) | 0.034**<br>(0.016) | 0.142<br>(0.193) | 0.162<br>(0.197)    |
| Unemp exp             |                      |                  |                    |                  | 0.054<br>(0.172)    |
| P9: P(Lose)           |                      |                  |                    |                  | 0.856***<br>(0.228) |
| N                     | 20932                | 12256            | 12254              | 587              | 586                 |

Notes: Difference-in-differences using Rep-won  $\times$  Post (Nov 2016) for non-swing states, 2016–2017. Two-way fixed effects (state, month) with controls; standard errors clustered by state. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Democrat-won states relative to Republican-won states, but the estimate is imprecise, reflecting the low frequency of the LMM and the limited number of state-level clusters. I therefore interpret this exercise as an illustration of belief dispersion rather than a definitive estimate of the causal effect of macro beliefs on search.

**Overall interpretation of robustness evidence.** Across specifications, three patterns are robust:

1. The perceived job-loss probability consistently predicts higher search intensity.
2. The effect survives demanding geographic and sectoral fixed effects.
3. The relationship is present both cross-sectionally and, to the extent permitted by the short panel, in within-person specifications (which are imprecisely estimated due to limited LMM repeats).

These exercises do not establish causal identification of belief shocks. They instead demonstrate that the empirical relationship linking perceived job-loss risk to search behavior is not easily explained by local shocks, industry conditions, functional-form choices, or fixed individual heterogeneity.

Taken together with the validation evidence in Section 2.4 (beliefs predict subsequent separations) and the mechanism evidence in Section 2.5 (stated motives and interaction patterns), the robustness exercises strengthen the interpretation that employed workers adjust search effort in response to perceived separation risk.

## C Model Appendix

This appendix collects calibration details, the full parameter list, derivations, and robustness exercises for the quantitative model.

### C.1 Calibration Details

**Simulated-regression check.** The main calibration targets the analytical two-state analogue of the empirical search semi-elasticity. I also solve the model, simulate worker-level observations, and run the same no-controls and controlled search regressions on model-generated data. This simulated-regression exercise is useful, but I do not use it as the headline target. In the current SCE calibration, the controlled simulated-regression coefficient is 2.60, above the empirical controlled coefficient of 1.69. A grid over  $\gamma$  can match

**Table A15.** Externally Set Parameters

| Parameter             | Description                                      | Value      | Source                                                          |
|-----------------------|--------------------------------------------------|------------|-----------------------------------------------------------------|
| $\beta$               | Monthly discount factor                          | 0.9967     | 4% annual rate                                                  |
| $b$                   | Flow value of unemployment                       | 0.40       | UI replacement benchmark                                        |
| $\alpha$              | Matching elasticity with respect to unemployment | 0.50       | <a href="#">Petrongolo and Pissarides (2001)</a>                |
| $\phi$                | Worker bargaining share                          | 0.50       | Hosios benchmark                                                |
| $\eta$                | Employed/unemployed contact ratio                | 0.10       | Employed search has lower time intensity                        |
| $\delta(B)/\delta(G)$ | Recession-expansion job-loss probability ratio   | 1.67       | Layoffs and discharges; <a href="#">Fujita and Ramey (2009)</a> |
| $y(G), y(B)$          | Productivity                                     | 1.00, 0.95 | 5% output gap                                                   |
| $\Pi_{GG}, \Pi_{BB}$  | State persistence                                | 0.99, 0.90 | NBER durations                                                  |
| $m(G), m(B)$          | Belief multipliers                               | 1.00, 1.00 | No state-dependent belief wedge                                 |

*Notes:* Parameters are monthly. The job-loss probability ratio governs cyclical; the level of job-loss risk is internally calibrated after accounting for the fraction of offers usable as insurance.

the simulated-regression coefficient, but doing so implies a less elastic structural search response than the analytical target. I therefore treat the simulated-regression exercise as a check on interpretation. It shows that the mapping from model effort to survey hours matters, while the qualitative conclusion remains the same: beliefs generate a meaningful search response at the worker level and a small effect on aggregate unemployment.

The additional moments in Table A16 show where the model fits the data and where it does not. In the SCE, employed search rises from 3.71 to 3.87 hours in high-slack months, a 4.4 percent increase. The model produces the same direction and a slightly larger response, 6.4 percent. Unemployment and job finding also move in the right direction. The main miss is total E-to-E mobility: E-to-E mobility is procyclical in the SCE, while insured-separation moves make total E-to-E mobility slightly higher in the bad state of the model.

Appendix Table A18 lists the parameter values used in the quantitative exercises. Parameters are monthly unless otherwise noted.

## C.2 Some Model Derivations

This appendix collects short derivations that clarify (i) the model's within-month timing, (ii) the search first-order condition when the worker evaluates unemployment risk using *perceived* job-loss probabilities, and (iii) the benchmark Nash wage rule used in the

**Table A16.** Untargeted Fit Diagnostics

| Moment                                             | Data                      | Model                    | Interpretation                                                               |
|----------------------------------------------------|---------------------------|--------------------------|------------------------------------------------------------------------------|
| Employed search cyclical-ity, $s(B)/s(G) - 1$      | +4.4%: 3.71 to 3.87 hours | +6.4%: 3.022 to 3.214    | Search rises in high-slack months and in the bad state.                      |
| Unemployment recession gap, $u(B) - u(G)$          | +2.52 pp: 5.75% to 8.27%  | +3.71 pp: 5.39% to 9.11% | The model recession is somewhat larger but in the empirical range.           |
| U-to-E cyclical-ity, $p_{UE}(B) - p_{UE}(G)$       | -0.87 pp: 20.5% to 19.6%  | -1.48 pp: 28.2% to 26.7% | Job finding falls in high-slack months.                                      |
| E-to-U inflow cyclical-ity, $p_{EU}(B)/p_{EU}(G)$  | 1.12: 1.10% to 1.23%      | 1.66: 1.61% to 2.67%     | The model has stronger inflow cyclical-ity than raw SCE transitions.         |
| Total E-to-E cyclical-ity, $p_{EE}(B) - p_{EE}(G)$ | -4.34 pp: 5.95% to 1.61%  | +0.08 pp: 3.83% to 3.91% | This is the clearest miss: total E-to-E mobility is procyclical in the data. |

*Notes:* These moments are not used to estimate the SCE calibration. The data column splits SCE worker-months into high- and low-slack months, with high slack defined as months above the median monthly SCE unemployment share. The search row uses employed respondents with observed search hours. The transition rows use next-wave SCE transitions; the E-to-E row is the all-employed lower-bound probability, coding missing transition-supplement responses as no E-to-E transition. The model column compares the recession and expansion states in the SCE calibration. The model's average U-to-E probability is calibrated to the stock-flow object rather than to the raw next-wave SCE transition probability, so the U-to-E row should be read as a cyclical comparison rather than a level target.

**Table A17.** Alternative Calibrations

| Profile              | $\rho_I$ | $\rho_J$ | $\omega_J$ | $s(G)$ | $s(B)$ | E-to-E share |
|----------------------|----------|----------|------------|--------|--------|--------------|
| SCE core             | 0.532    | 0.108    | 0.145      | 3.022  | 3.214  | 0.094        |
| Higher insurance use | 0.750    | 0.102    | 0.146      | 2.089  | 2.282  | 0.146        |
| CPS                  | 0.223    | 0.079    | 0.053      | 1.933  | 2.127  | 0.050        |

*Notes:* The higher-insurance-use profile fixes  $\rho_I = 0.75$  and recalibrates the remaining parameters. The CPS profile uses lower E-to-E and wage-gain targets. All profiles match the perceived-risk and search-response anchors.



**Table A18.** Full Parameter List

| Parameter              | Description                                   | Value        |
|------------------------|-----------------------------------------------|--------------|
| $\beta$                | Discount factor                               | 0.9967       |
| $\phi$                 | Worker bargaining weight                      | 0.5          |
| $\alpha$               | Matching elasticity (unemployment)            | 0.5          |
| $b$                    | Flow value of unemployment                    | 0.4          |
| $y(G), y(B)$           | Productivity levels                           | 1.00, 0.95   |
| $\Pi_{GG}, \Pi_{BB}$   | Markov persistence (monthly)                  | 0.99, 0.90   |
| $\delta(G), \delta(B)$ | Job-loss probabilities (monthly)              | 0.024, 0.039 |
| $\eta$                 | Employed/unemployed contact probability ratio | 0.1          |
| $\chi$                 | Matching efficiency                           | 0.338        |
| $\kappa$               | Vacancy posting cost                          | 0.508        |
| $\psi$                 | Search effectiveness                          | 26.282       |
| $\gamma$               | Search cost curvature                         | 0.0229       |
| $\zeta$                | Belief level shifter                          | 0.775        |

*Notes:* Parameters are monthly unless stated otherwise. The transition matrix has off-diagonal elements  $\Pi_{GB} = 1 - \Pi_{GG}$  and  $\Pi_{BG} = 1 - \Pi_{BB}$ . Productivity is normalized to  $y(G) \approx 1$ . The baseline sets  $m(G) = m(B) = 1$ , so the cyclicalty of perceived risk is inherited from objective separation risk rather than belief distortions.

quantitative implementation.

Throughout, time is monthly and the aggregate state follows  $z \in \{G, B\}$  with transition matrix  $\Pi(z, z')$ .

### C.3 Monthly probabilities and within-month timing

Let  $\delta(z)$  denote the *objective* monthly probability that the current job ends in state  $z$ :

$$\delta(z) \in (0, 1), \quad \text{with survival probability } 1 - \delta(z). \quad (12)$$

Employed search effort  $s$  maps into an offer probability

$$p_{\text{offer}}(z, s) = 1 - \exp\{-\eta\chi\theta(z)^{1-\alpha}\psi g(s)\}. \quad (13)$$

**Ordering within the month.** The implementation follows the timing in Section 3.1: the current aggregate state  $z$  is realized, workers choose search, offers and separations are realized within the month in that same state, and then the economy transitions to  $z'$  for continuation values. With this ordering, the *objective* probability of ending the month

unemployed, given current state  $z$  and effort  $s$ , is

$$\delta_{EU}(z, s) = (1 - p_{\text{offer}}(z, s)) \delta(z). \quad (14)$$

Equivalently, offers reduce unemployment inflows by reducing the probability that a separation “hits” a worker who lacks a fallback option.

#### C.4 Perceived separation risk and perceived employment-to-unemployment probability

Workers may perceive job-loss risk differently from the objective probability. Let the perceived monthly job-loss probability be

$$\tilde{\delta}(z) = \zeta \cdot m(z) \cdot \delta(z), \quad (15)$$

where  $\zeta > 0$  is a level shifter and  $m(z)$  allows perceived risk to vary differently across aggregate states.

Under the same within-month ordering (search and offer realization within month, with separation risk realized at month-end), the worker’s *perceived* probability of entering unemployment in current state  $z$  is

$$\tilde{\delta}_{EU}(z, s) = (1 - p_{\text{offer}}(z, s)) \tilde{\delta}(z). \quad (16)$$

This is the object that enters the worker continuation value in the implementation.

#### C.5 Deriving the search first-order condition

Fix current aggregate state  $z$  and let the worker choose effort  $s(z) \equiv s$  after observing  $z$ . Let  $W(z)$  and  $U(z)$  denote the worker values, computed under the worker’s subjective transition probabilities only through  $\tilde{\delta}_{EU}$ .

The employed value satisfies

$$W(z) = w(z) - c(s) + \beta \sum_{z'} \Pi(z, z') \left[ (1 - \tilde{\delta}_{EU}(z, s)) W(z') + \tilde{\delta}_{EU}(z, s) U(z') \right]. \quad (17)$$

Differentiate (17) with respect to  $s$  (holding  $w(\cdot)$ ,  $\theta(\cdot)$ , and the continuation values

$W(\cdot), U(\cdot)$  fixed from the perspective of the individual worker's static choice):

$$\begin{aligned} 0 &= -c'(s) + \beta \sum_{z'} \Pi(z, z') \left[ -\frac{\partial \tilde{\delta}_{EU}(z, s)}{\partial s} W(z') + \frac{\partial \tilde{\delta}_{EU}(z, s)}{\partial s} U(z') \right] \\ &= -c'(s) + \beta \left[ -\frac{\partial \tilde{\delta}_{EU}(z, s)}{\partial s} \right] \sum_{z'} \Pi(z, z') (W(z') - U(z')). \end{aligned} \quad (18)$$

Using (16), and noting that  $\tilde{\delta}(z)$  does not depend on  $s$ ,

$$\frac{\partial \tilde{\delta}_{EU}(z, s)}{\partial s} = -\tilde{\delta}(z) \frac{\partial p_{\text{offer}}(z, s)}{\partial s}. \quad (19)$$

Substitute (19) into (18) to obtain the search first-order condition:

$$c'(s(z)) = \beta \tilde{\delta}(z) \frac{\partial p_{\text{offer}}(z, s(z))}{\partial s} \sum_{z'} \Pi(z, z') (W(z') - U(z')). \quad (20)$$

Finally, with  $p_{\text{offer}}(z, s) = 1 - \exp\{-\xi(z, s)\}$  and  $\xi(z, s) = \lambda_e(\theta(z))\psi g(s)$ ,

$$\frac{\partial p_{\text{offer}}(z, s)}{\partial s} = \exp(-\xi(z, s)) \lambda_e(\theta(z))\psi g'(s) \geq 0. \quad (21)$$

**Economic interpretation.** Equation (20) makes the mechanism transparent. Search is attractive when (i) perceived job-loss risk  $\tilde{\delta}(z)$  is high, (ii) effort moves offer availability (high  $\partial p_{\text{offer}}/\partial s$ ), and (iii) the expected continuation surplus,  $\sum_{z'} \Pi(z, z') (W(z') - U(z'))$ , is large.

## C.6 Benchmark Nash wage with free entry

This subsection derives the benchmark wage rule used in the quantitative implementation. The key modeling choice is that wages are set as in a standard DMP environment and do not renegotiate within the month in response to the worker's search choice. Employed search affects flows (through (14)) but not the wage schedule.

**Firm value of a filled job.** Let  $J(z)$  denote the firm's value of a filled job in state  $z$ , under the objective separation process. Because separations are realized within the current month in current state  $z$ , the firm Bellman equation is

$$J(z) = y(z) - w(z) + \beta (1 - \delta(z)) \sum_{z'} \Pi(z, z') J(z'). \quad (22)$$

**Free entry.** Let  $V(z)$  be the value of a vacancy posted in state  $z$ . Free entry implies  $V(z) = 0$ . The monthly filling probability is  $p_f(z) = 1 - \exp\{-\chi\theta(z)^{-\alpha}\}$ . Posting a vacancy costs  $\kappa$  for the month; if the vacancy is filled, the firm obtains the continuation value of a filled job next month. The zero-profit condition is therefore

$$\kappa = \beta p_f(z) \sum_{z'} \Pi(z, z') J(z'), \quad \text{with } p_f(z) = 1 - \exp\{-\chi\theta(z)^{-\alpha}\}. \quad (23)$$

**Nash bargaining.** Let the worker and firm split the match surplus according to Nash bargaining with worker share  $\phi \in (0, 1)$ . Let the worker outside option be unemployment, with flow value  $b$ . Under the benchmark wage protocol, the worker's outside option is summarized by  $b$  and tightness; employed search affects unemployment risk but does not enter the bargaining problem directly.

Under standard DMP bargaining with free entry, the wage can be written in the familiar reduced form:

$$w(z) = (1 - \phi) b + \phi(y(z) + \kappa \theta(z)). \quad (24)$$

**Connection to discrete-time exact probabilities.** In discrete time, the term  $\kappa\theta(z)$  corresponds to the expected hiring cost that the worker saves the firm by accepting a match. The discrete-time analogue replaces  $\theta(z)$  by the ratio of worker and vacancy transition probabilities,

$$\kappa \theta(z) \approx \kappa \frac{p_u(z)}{p_f(z)}, \quad (25)$$

where  $p_u(z) = 1 - \exp\{-\chi\theta(z)^{1-\alpha}\}$  and  $p_f(z) = 1 - \exp\{-\chi\theta(z)^{-\alpha}\}$ . This keeps the wage expression aligned with the monthly transition probabilities used elsewhere in the model.

## C.7 Insurance-off algebraic comparison

The insurance channel is the difference between the unemployment inflow probability with and without offers being usable at separation time. Using (14) and the insurance-off counterpart  $\delta_{EU}^{\text{off}}(z) = \delta(z)$ , we have

$$\delta_{EU}^{\text{off}}(z) - \delta_{EU}(z, s) = \delta(z) p_{\text{offer}}(z, s). \quad (26)$$

Thus, the insurance value of search is larger when separations are more likely (high  $\delta$ ) and when offers are more likely to be in hand (high  $p_{\text{offer}}$ ).

## C.8 Proofs for Section 3.6

This appendix gives the short arguments behind the analytical implications in Section 3.6. The logic comes from two objects: the search first-order condition (6) and the mapping from offers to unemployment transitions (7).

*Proof of Proposition 1.* Fix  $z$  and hold tightness, continuation values, and ladder values fixed. Let

$$A(z) \equiv \sum_{z'} \Pi(z, z') \rho_J \Delta_J(z'), \quad B(z) \equiv \sum_{z'} \Pi(z, z') \rho_I (W(z') - U(z')).$$

Define

$$F(s, \tilde{\delta}(z)) \equiv c'(s) - \beta p_{\text{offer},s}(z, s) [(1 - \tilde{\delta}(z))A(z) + \tilde{\delta}(z)B(z)].$$

An interior optimum satisfies  $F(s(z), \tilde{\delta}(z)) = 0$ . By (4),  $p_{\text{offer},s}(z, s) > 0$ , so

$$F_{\tilde{\delta}} = -\beta p_{\text{offer},s}(z, s) [B(z) - A(z)] < 0.$$

The inequality uses the condition in the proposition,  $B(z) > A(z)$ . Also,

$$p_{\text{offer},ss}(z, s) = \exp\left(-\lambda_e(\theta(z))\psi g(s)\right) \lambda_e(\theta(z))\psi \left[g''(s) - \lambda_e(\theta(z))\psi (g'(s))^2\right] < 0,$$

because  $g'(s) > 0$ ,  $g''(s) \leq 0$ , and  $\lambda_e(\theta(z))\psi > 0$ . Hence

$$F_s = c''(s) - \beta p_{\text{offer},ss}(z, s) [(1 - \tilde{\delta}(z))A(z) + \tilde{\delta}(z)B(z)] > 0.$$

By the implicit function theorem,

$$\frac{\partial s(z)}{\partial \tilde{\delta}(z)} = -\frac{F_{\tilde{\delta}}}{F_s} > 0.$$

□

*Proof of Proposition 2.* From (7),  $\delta_{EU}(z, s) = \delta(z)(1 - \rho_I p_{\text{offer}}(z, s))$ . Because  $p_{\text{offer}}(z, s)$  is strictly increasing in  $s$ ,  $\delta_{EU}(z, s)$  is strictly decreasing in  $s$ . For a given offer probability,  $\delta_{EU}(z, s)$  is increasing in the objective job-loss probability  $\delta(z)$ . □

*Proof of Proposition 3.* Let

$$\mathcal{MB}(z; s) = p_{\text{offer},s}(z, s) \sum_{z'} \Pi(z, z') [(1 - \tilde{\delta}(z))\rho_J \Delta_J(z') + \tilde{\delta}(z)\rho_I (W(z') - U(z'))].$$

Define  $F_z(s) \equiv c'(s) - \beta \mathcal{MB}(z; s)$ . Under Assumption 1,  $c'(s)$  is strictly increasing. Since  $p_{\text{offer},ss}(z, s) < 0$  under the maintained search technology,

$$\frac{\partial \mathcal{MB}(z; s)}{\partial s} = p_{\text{offer},ss}(z, s) \sum_{z'} \Pi(z, z') [(1 - \tilde{\delta}(z)) \rho_J \Delta_J(z') + \tilde{\delta}(z) \rho_I (W(z') - U(z'))] < 0.$$

Therefore,

$$F'_z(s) = c''(s) - \beta \frac{\partial \mathcal{MB}(z; s)}{\partial s} > 0,$$

so  $F_z(s)$  is strictly increasing and has a unique root.

If  $\mathcal{MB}(B; s) \geq \mathcal{MB}(G; s)$  in a neighborhood containing the relevant effort choices, then  $F_B(s) \leq F_G(s)$  in that neighborhood. Since  $F_G(s(G)) = 0$ , it follows that  $F_B(s(G)) \leq 0$ . Because  $F_B$  is increasing and crosses zero at  $s(B)$ , we must have  $s(B) \geq s(G)$ .  $\square$

**Proposition 5** (UI crowds out precautionary on-the-job search). *Holding fixed  $\theta(\cdot)$ ,  $\tilde{\delta}(\cdot)$ , and  $\Pi$ , a higher unemployment benefit  $b$  reduces the optimal on-the-job search intensity  $s(z)$  for each aggregate state  $z$ .*

*Proof of Proposition 5.* Holding fixed  $\theta(\cdot)$ ,  $\tilde{\delta}(\cdot)$ , and  $\Pi$ , the only channel through which  $b$  affects the marginal benefit of search in (6) is the continuation surplus term  $\sum_{z'} \Pi(z, z') (W(z') - U(z'))$ . Under the maintained condition that  $\partial(W(z') - U(z'))/\partial b < 0$  for each  $z'$ , this term decreases in  $b$ , so the right-hand side of (6) decreases in  $b$ . Since  $c'(\cdot)$  is strictly increasing, the optimal choice  $s(z)$  must decrease with  $b$ . The statement about unemployment inflows follows from Proposition 2.  $\square$

*Proof of Proposition 4.* If  $s^{\text{base}}(B) > s^{\text{fix}}(B)$ , then by Proposition 2,  $\delta_{EU}(B, s^{\text{base}}(B)) < \delta_{EU}(B, s^{\text{fix}}(B))$ . Holding  $p_u(B)$  fixed, the flow-balance expression (9) is increasing in  $\delta_{EU}(B, \cdot)$ , so steady-state unemployment is lower under baseline search than under fixed search.  $\square$

## C.9 Counterfactual Details

Each reported number re-solves the state-contingent equilibrium under the relevant calibration and then computes the fixed-search recession comparison using recession-state contact probabilities.

**Table A19.** Counterfactual Details

| Calibration                           | Exercise                         | Dampening |
|---------------------------------------|----------------------------------|-----------|
| SCE core                              | Job-ladder calibration           | 0.046 pp  |
| Higher insurance use, $\rho_I = 0.75$ | More offers usable at separation | 0.112 pp  |
| CPS                                   | CPS calibration                  | 0.030 pp  |

*Notes:* Dampening is  $100 \times [u_B^{\text{fixed search}} - u_B^{\text{reoptimized}}]$ . The higher-insurance-use case fixes  $\rho_I = 0.75$  because self-reported job-loss E-to-E transitions plausibly understate the share of outside options that can protect workers at separation.

## C.10 AR(1) State-of-the-World Extension

As a check on the two-state aggregation, I also solve an extension in which the aggregate state is a discretized AR(1) state-of-the-world process,

$$\log y_t = \rho \log y_{t-1} + \varepsilon_t.$$

I use a seven-state Rouwenhorst approximation. The search and offer-arrival block is tied to the SCE/FMST evidence used in the main calibration (Faberman et al., 2022). The AR(1) process is a persistence robustness rather than a separately estimated productivity process. Job-loss risk is mapped into the state of the world so that a five percent decline in the state implies the same job-loss probability ratio as in the two-state calibration.

I recalibrate the AR(1) grid to the main average moments. The recalibration hits offer receipt and average E-to-E mobility closely, and comes close on unemployment and U-to-E flows. It does not fully recover the targeted search-belief semi-elasticity: the local AR(1) value is 1.38, compared with the target 1.69. Table A20 should therefore be read as a disciplined robustness exercise rather than a replacement for the main two-state calibration.

## C.11 Ladder Value Sensitivity

A natural concern is that the ladder value is understated. The headline calibration maps the observed wage gain from E-to-E moves into a one-period value gain. If these gains are persistent, this mapping may make ladder search too weak. Table A21 varies  $\omega_l$  by factors of two and five, holding the SCE core  $\rho_I$  and  $\rho_J$  fixed and re-solving the state-contingent equilibria.

Increasing the ladder value raises search in both states, especially in expansions, and reduces the aggregate stabilization attributed to precautionary re-optimization. This goes against the concern that the small aggregate effect is caused by an undervalued ladder

**Table A20.** AR(1) State-of-the-World Extension

| Object                     | Good state | Bad state | Difference |
|----------------------------|------------|-----------|------------|
| Productivity/state $y$     | 1.000      | 0.949     | -0.051     |
| Search effort $s$          | 0.477      | 0.510     | 0.033      |
| Offer probability          | 0.330      | 0.322     | -0.009     |
| U-to-E probability         | 0.282      | 0.262     | -0.020     |
| Unemployment               | 5.30%      | 9.23%     | 3.93 pp    |
| Unemployment, fixed search | –          | 9.33%     | –          |
| Dampening, PE              | –          | 0.098 pp  | 2.44%      |

*Notes:* The good state is the grid point with  $y = 1$ . The bad state is the grid point closest to a five percent decline. Fixed search holds bad-state employed search at the good-state level while keeping bad-state contact probabilities fixed, so the dampening entry is a partial-equilibrium comparison.

**Table A21.** Sensitivity to the Ladder Value

| $\omega_J$ multiplier | $\omega_J$ | $s(G)$ | $s(B)$ | $p_{\text{offer}}(G)$ | $p_{\text{offer}}(B)$ | Dampening          |
|-----------------------|------------|--------|--------|-----------------------|-----------------------|--------------------|
| 1                     | 0.145      | 3.022  | 3.214  | 0.330                 | 0.322                 | 0.046 pp (1.22%)   |
| 2                     | 0.290      | 3.841  | 3.923  | 0.366                 | 0.351                 | 0.016 pp (0.43%)   |
| 5                     | 0.726      | 5.630  | 5.551  | 0.428                 | 0.406                 | -0.010 pp (-0.30%) |

*Notes:* The sensitivity varies the ladder gain parameter holding the SCE core  $\rho_I$  and  $\rho_J$  fixed and re-solves the state-contingent equilibria. Larger ladder values make employed search less precautionary at the margin, so the fixed-search unemployment comparison shrinks.

motive. A stronger ladder motive makes total search larger but makes the marginal cyclical insurance channel smaller.

**Insurance-off benchmark.** In the job-ladder model, shutting down the insurance use of offers corresponds to setting  $\rho_I = 0$ . Then the realized employment-to-unemployment probability becomes

$$p_{EU}^{\text{off}}(z) = \delta(z),$$

independent of employed search. Ladder search may still be valuable through  $\rho_J$  and  $\omega_J$ , but it does not directly reduce unemployment inflows. This is why the aggregate unemployment margin in the main text is governed by  $\rho_I$ , while total E-to-E mobility is governed jointly by  $\rho_I$  and  $\rho_J$ .